



US012418593B2

(12) **United States Patent**
Canpolat et al.

(10) **Patent No.:** **US 12,418,593 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **MECHANISM TO ENABLE A FEDERATED ONBOARDING SERVICE IN AN OPENROAMING FRAMEWORK**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 610 days.

(21) Appl. No.: **17/645,974**

(22) Filed: **Dec. 24, 2021**

(65) **Prior Publication Data**

US 2022/0124167 A1 Apr. 21, 2022

Related U.S. Application Data

(60) Provisional application No. 63/246,590, filed on Sep. 21, 2021.

(51) **Int. Cl.**
H04L 67/306 (2022.01)
H04L 67/14 (2022.01)

(52) **U.S. Cl.**
CPC **H04L 67/306** (2013.01); **H04L 67/14** (2013.01)

(58) **Field of Classification Search**

CPC H04L 67/306; H04L 67/14; H04L 67/12; H04W 8/12; H04W 12/033; H04W 12/069; H04W 12/35

See application file for complete search history.

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Primary Examiner — Soe Hlaing

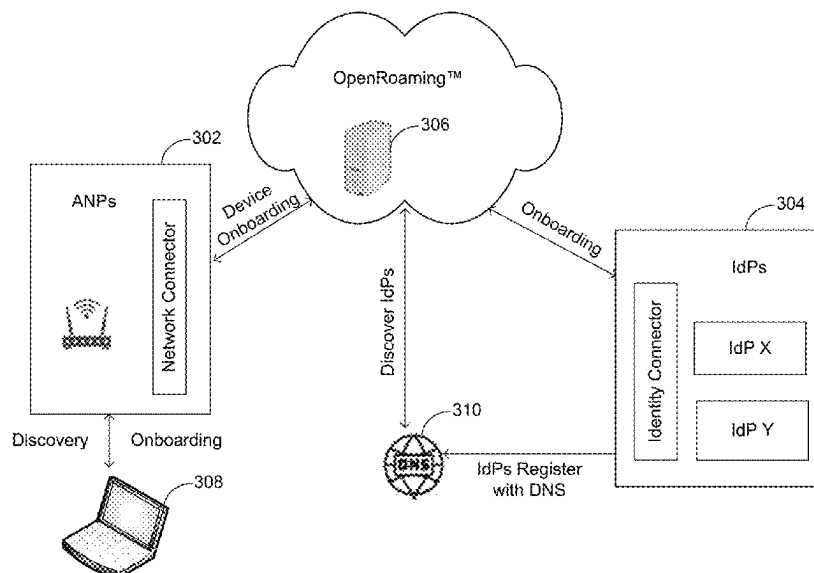
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(57) **ABSTRACT**

This disclosure describes systems, methods, and devices related to a mechanism to enable a federated onboarding service in an OpenRoaming™ framework. A device may receive a prompt to initiate onboarding of the device to a framework. The device may the initiate a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework. The device may further receive a list of available identity providers (IdPs) configured for the framework and receive a selection of an IdP of the list of available IdPs. The device may then establish a connection between the IdP and the device. The device may receive an IdP-generated user-specific profile, and the device may connect to the framework using the user-specific profile.

20 Claims, 11 Drawing Sheets

← 300



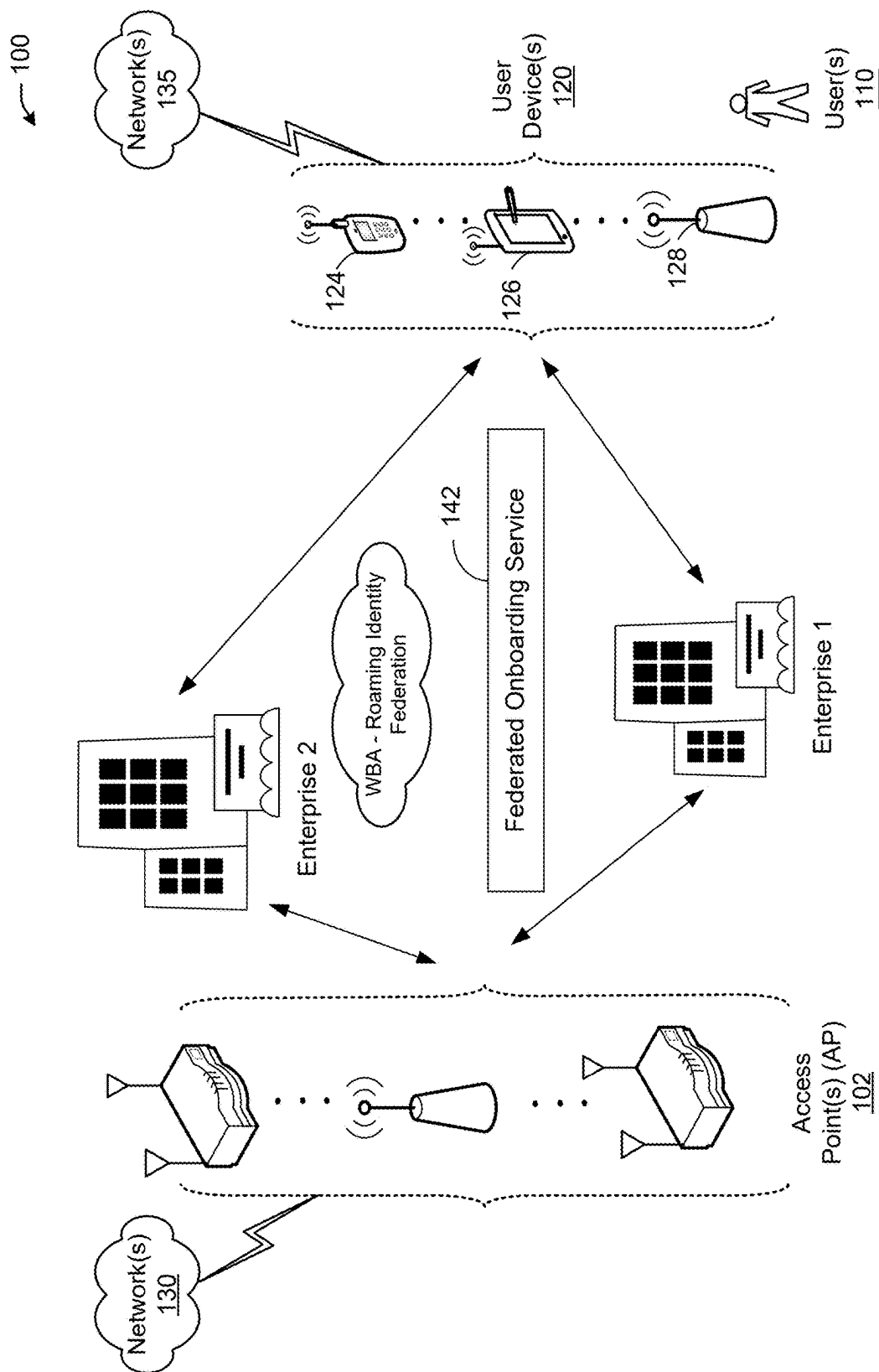


FIG. 1

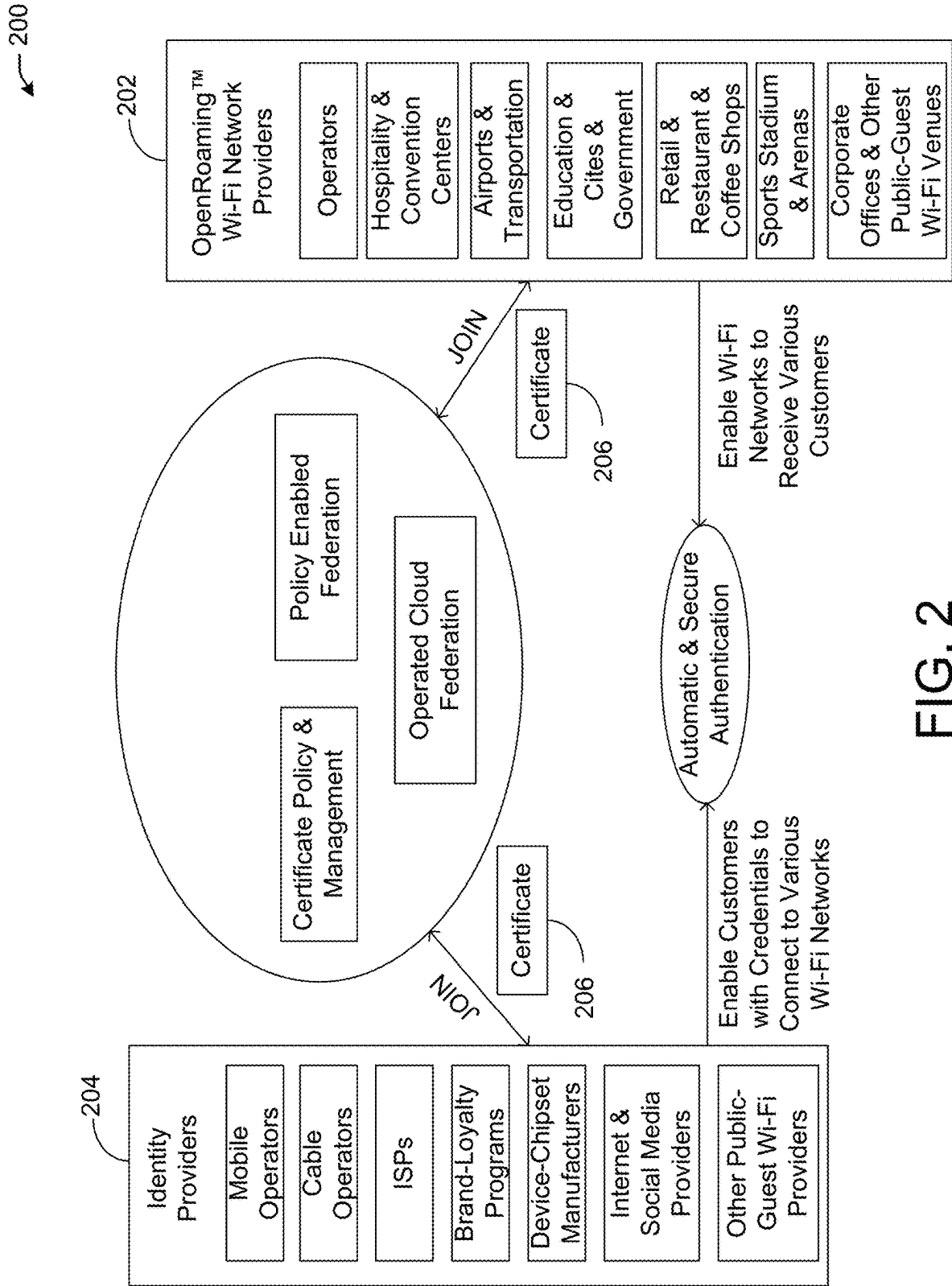


FIG. 2

300

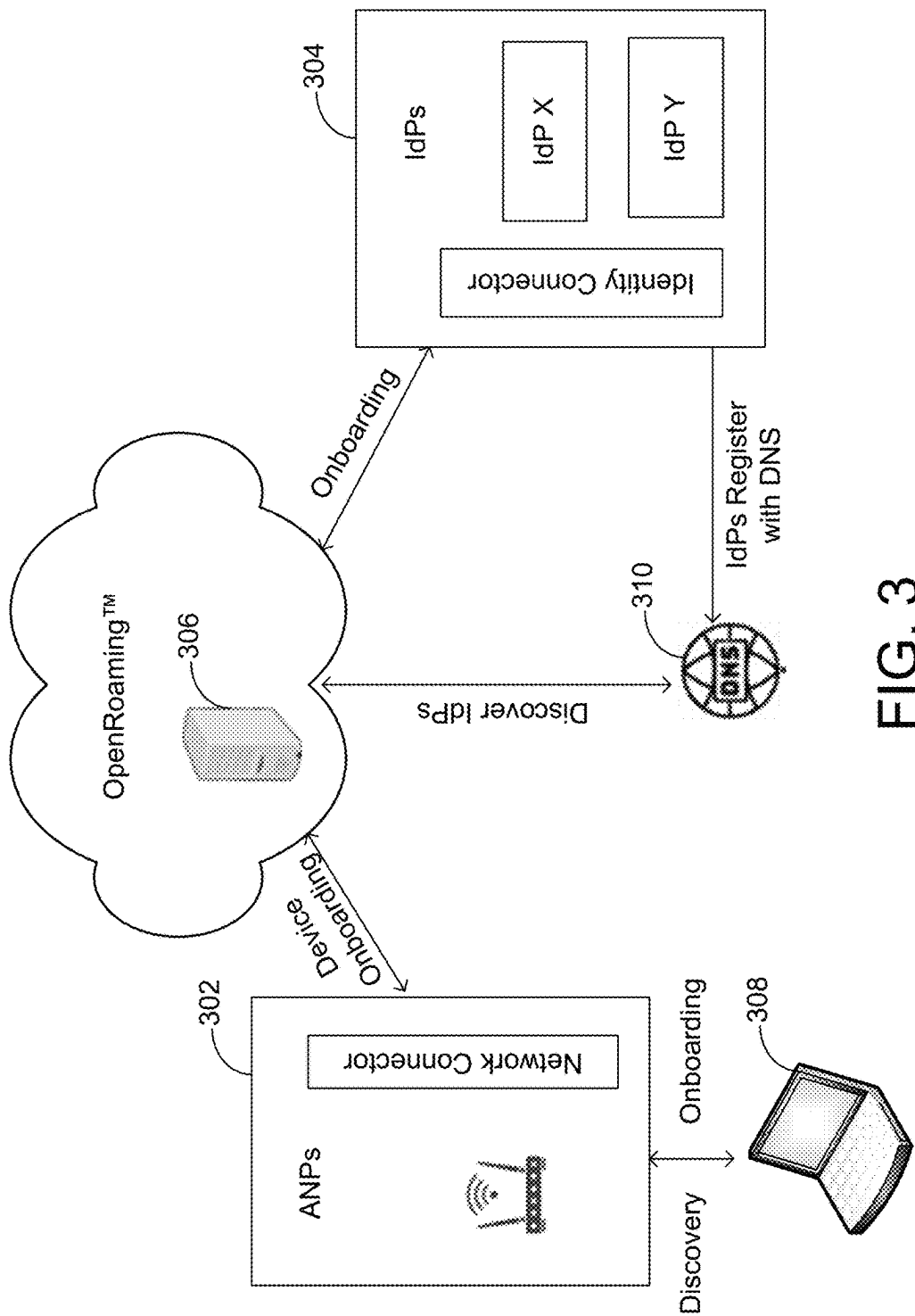


FIG. 3

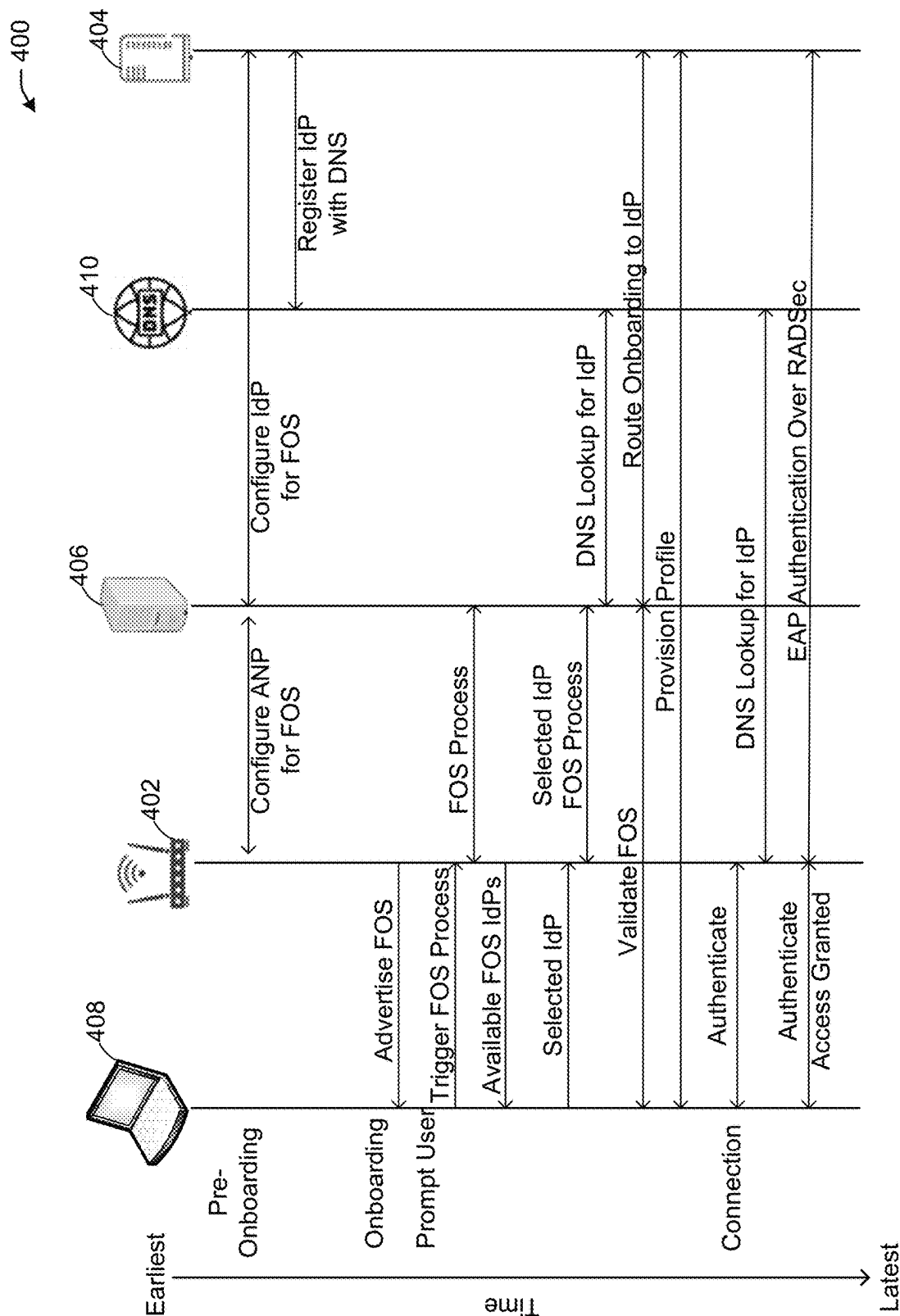


FIG. 4

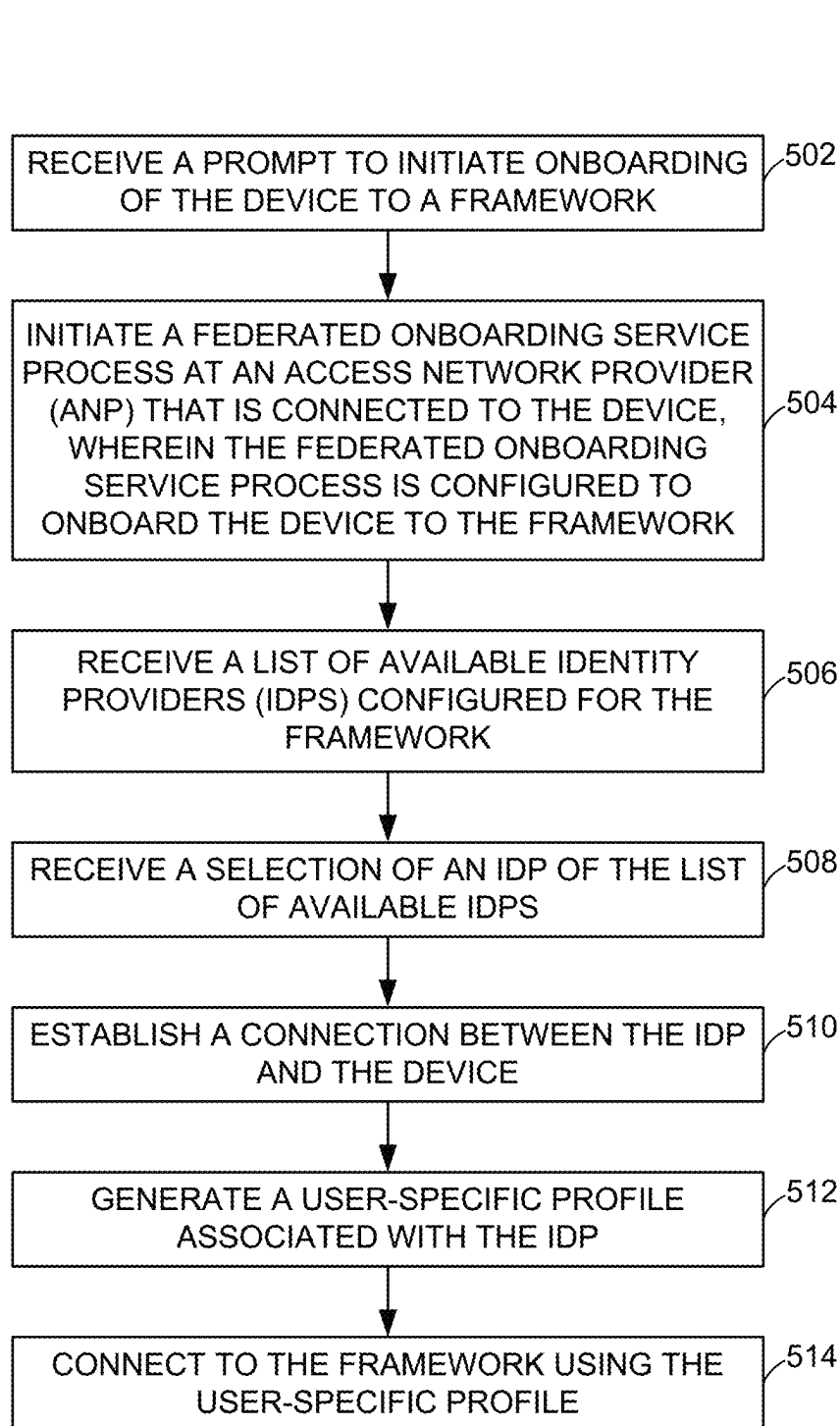


FIG. 5

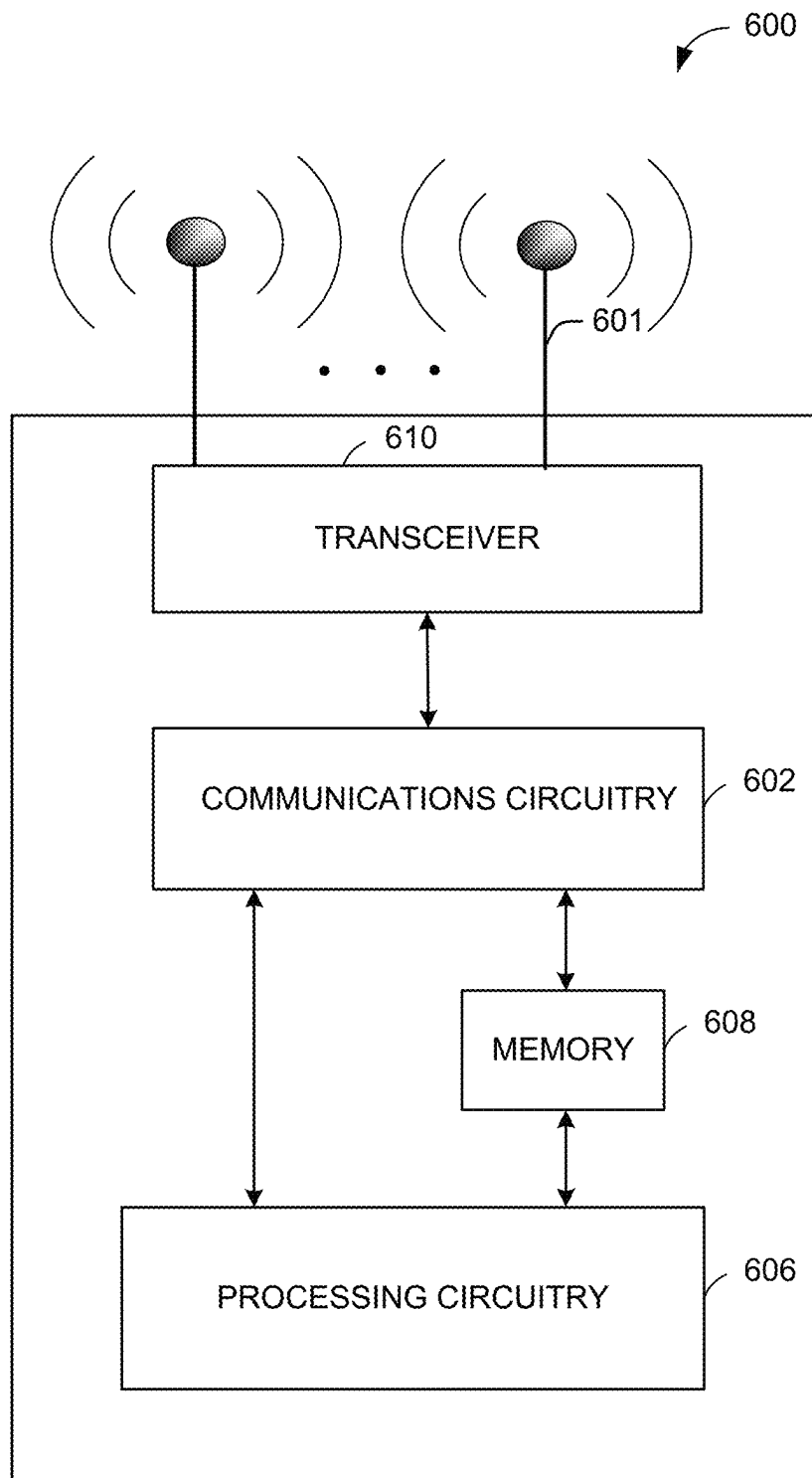


FIG. 6

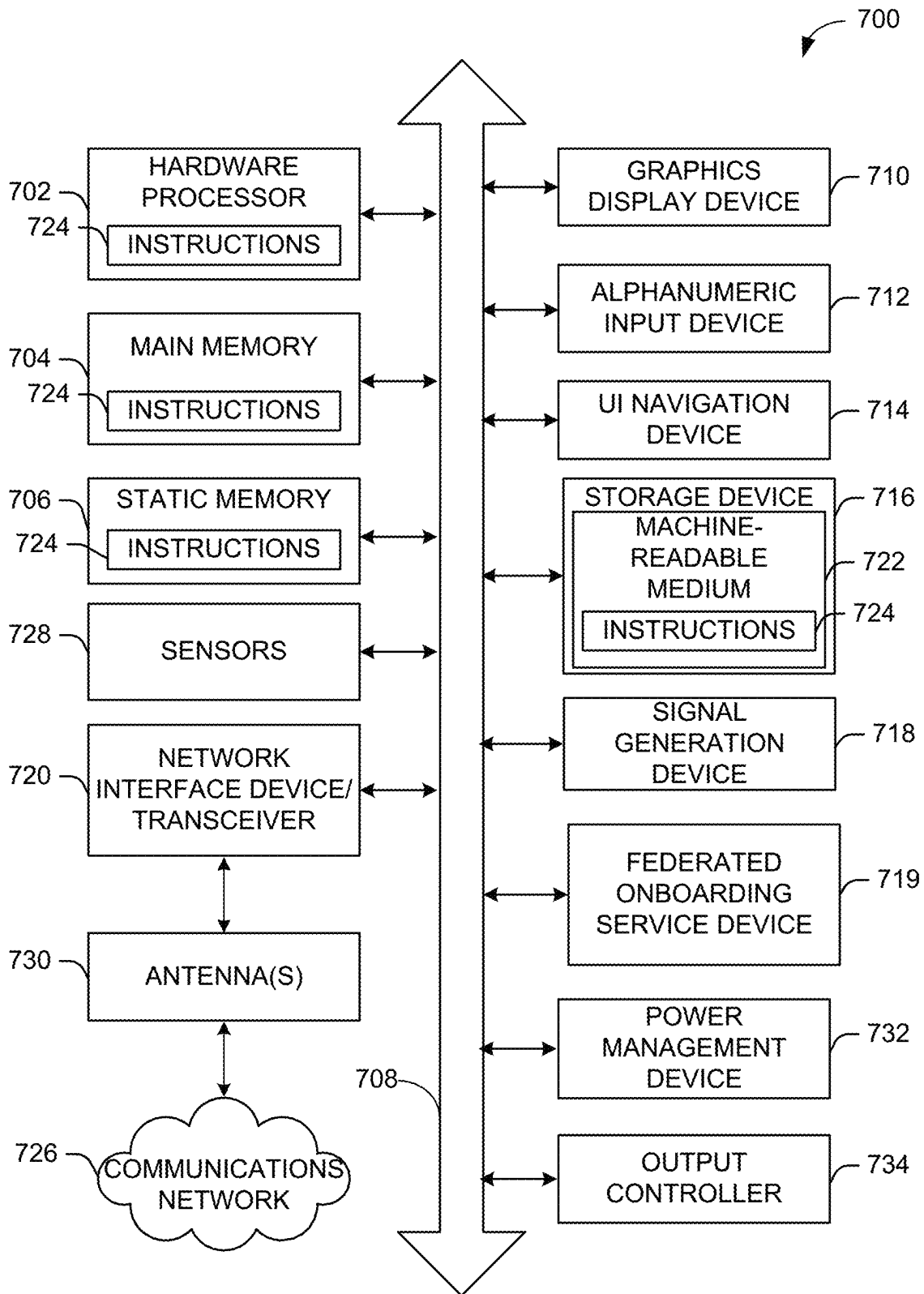


FIG. 7

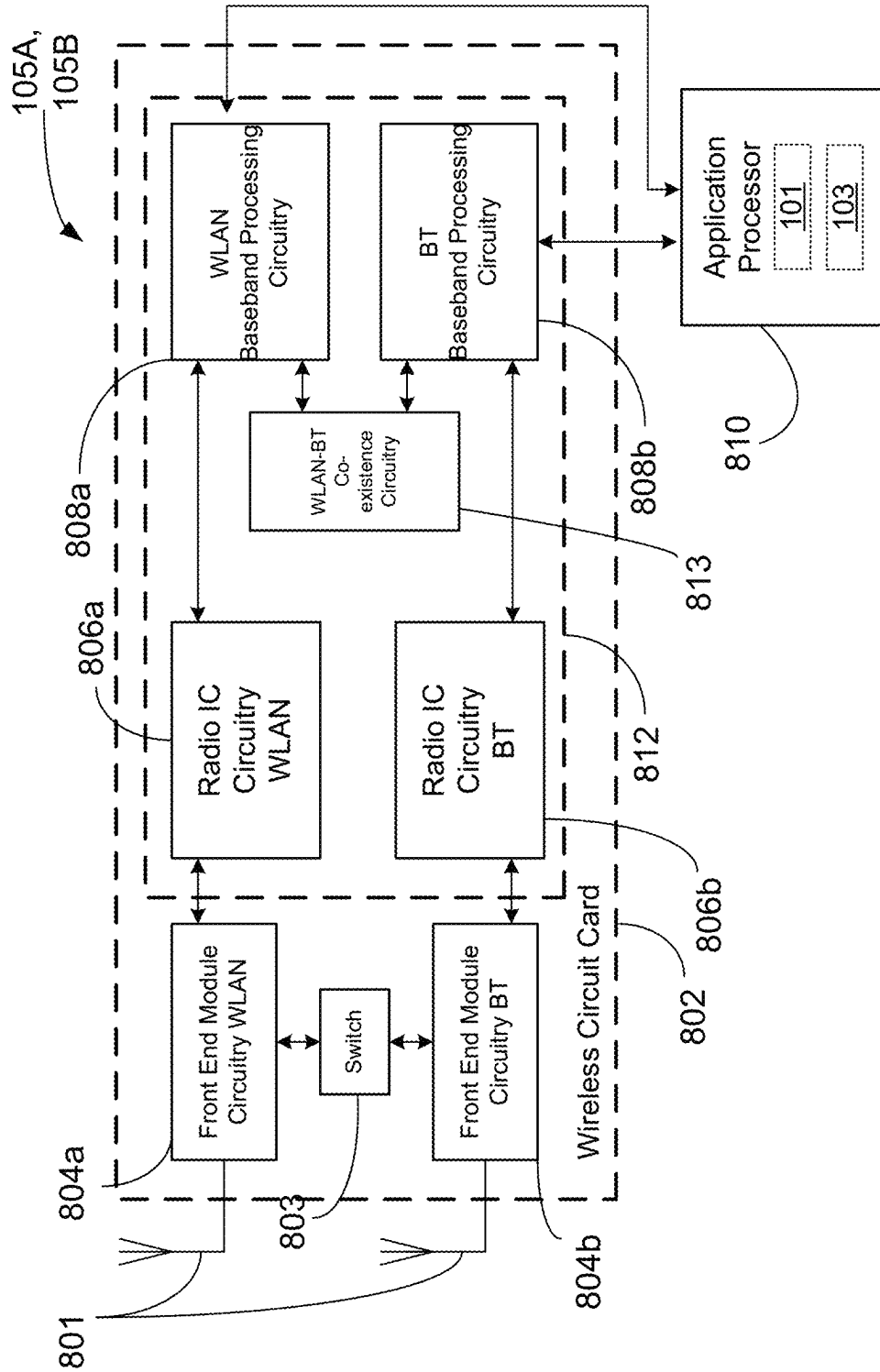


FIG. 8

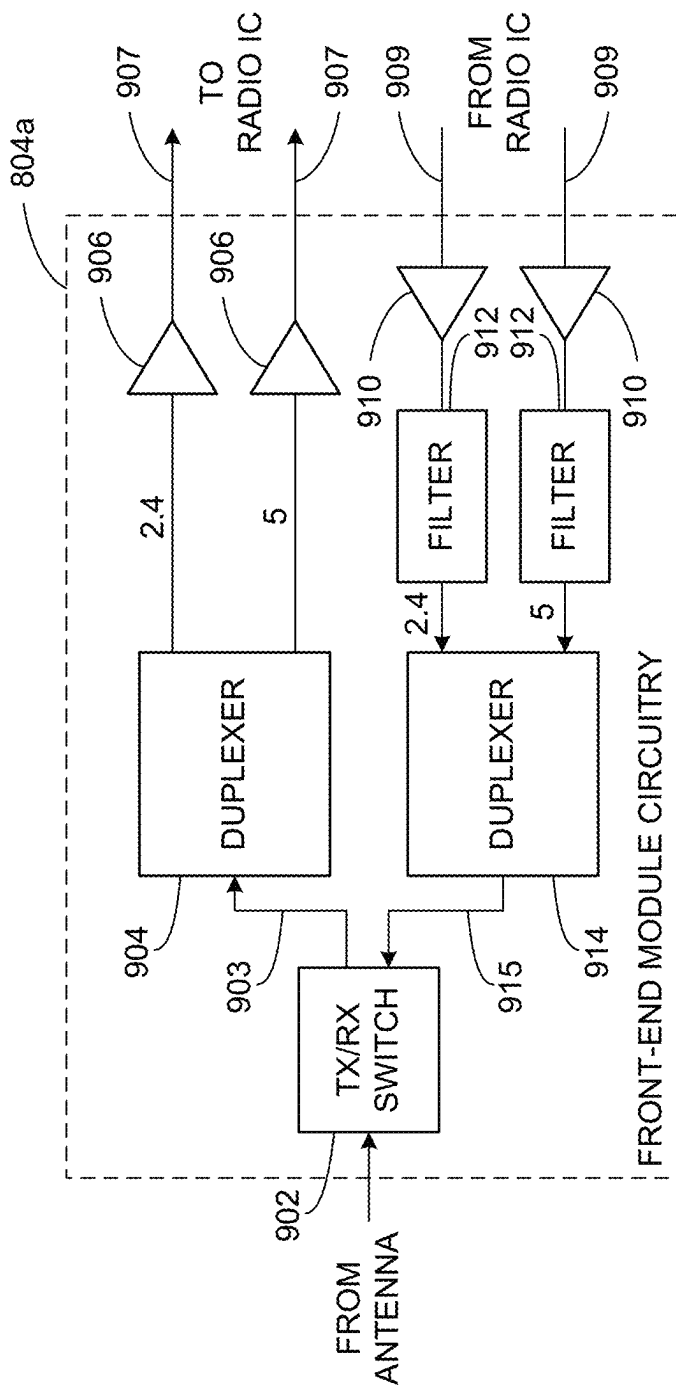


FIG. 9

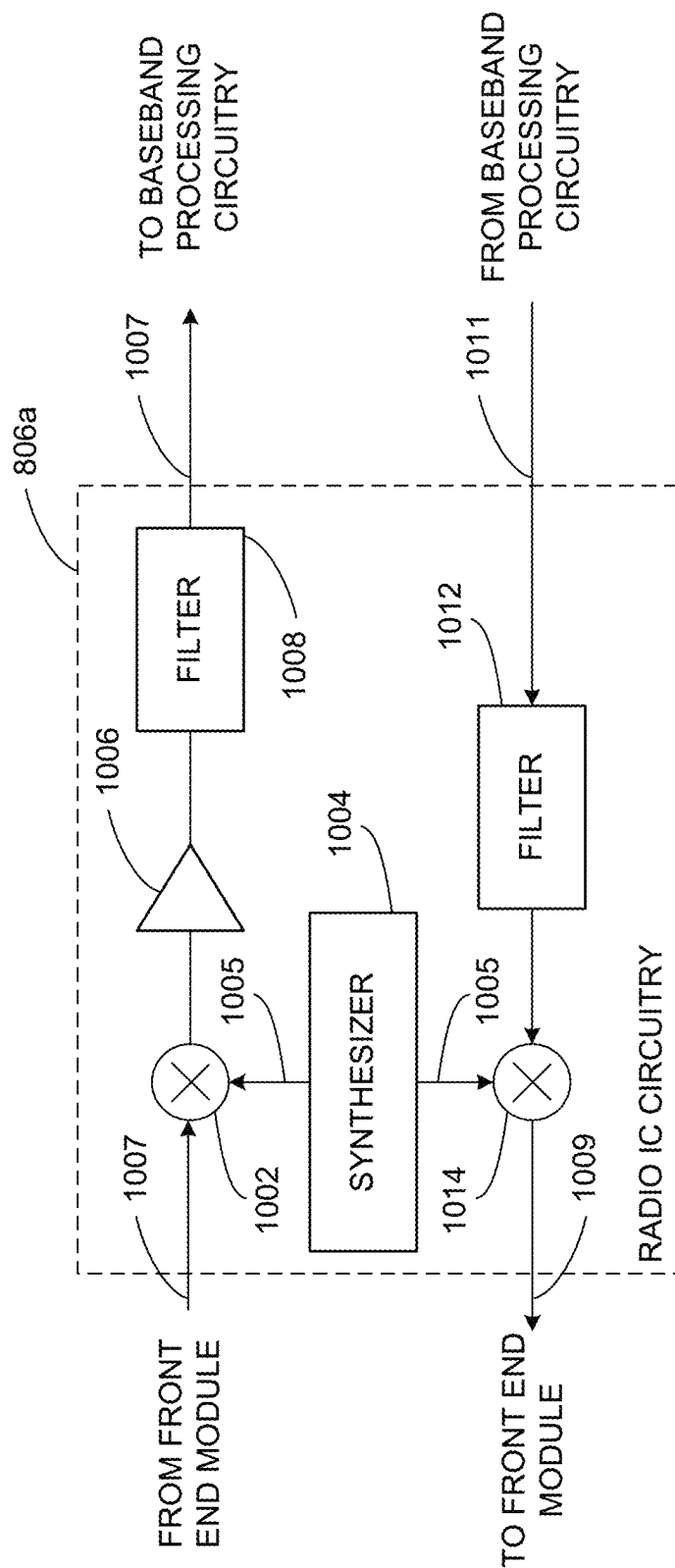


FIG. 10

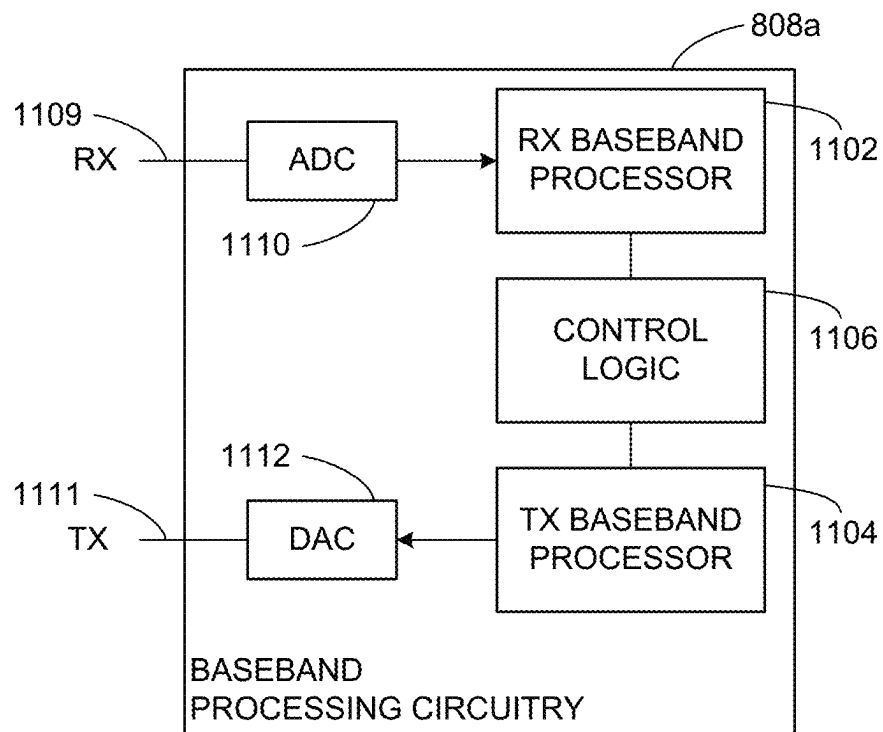


FIG. 11

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MECHANISM TO ENABLE A FEDERATED ONBOARDING SERVICE IN AN OPENROAMING FRAMEWORK

CROSS-REFERENCE TO RELATED PATENT
APPLICATION(S)

This application claims the benefit of U.S. Provisional Application No. 63/246,590, filed Sep. 21, 2021, the disclosure of which is incorporated by reference as set forth in full.

TECHNICAL FIELD

This disclosure generally relates to systems and methods for wireless communications and, more particularly, to a mechanism to enable a federated onboarding service in an OpenRoaming framework.

BACKGROUND

Wireless Broadband Alliance (WBA) has launched Federated OpenRoaming solution that enables creation of One-Global Wi-Fi network by providing PKI based security and roaming framework among Access Network Providers (ANPs) and Id Providers (IdPs). However, onboarding of devices on to OpenRoaming networks is still a challenge and here we provide Federated Onboarding Services that can be enabled within OpenRoaming framework.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a network diagram illustrating an example network environment for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

FIG. 2 depicts an illustrative schematic diagram for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

FIG. 3 depicts an illustrative schematic diagram for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

FIG. 4 depicts an illustrative schematic diagram for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

FIG. 5 illustrates a flow diagram of illustrative process for an illustrative federated onboarding service system, in accordance with one or more example embodiments of the present disclosure.

FIG. 6 illustrates a functional diagram of an exemplary communication station that may be suitable for use as a user device, in accordance with one or more example embodiments of the present disclosure.

FIG. 7 illustrates a block diagram of an example machine upon which any of one or more techniques (e.g., methods) may be performed, in accordance with one or more example embodiments of the present disclosure.

FIG. 8 is a block diagram of a radio architecture in accordance with some examples.

FIG. 9 illustrates an example front-end module circuitry for use in the radio architecture of FIG. 8, in accordance with one or more example embodiments of the present disclosure.

FIG. 10 illustrates an example radio IC circuitry for use in the radio architecture of FIG. 8, in accordance with one or more example embodiments of the present disclosure.

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FIG. 11 illustrates an example baseband processing circuitry for use in the radio architecture of FIG. 8, in accordance with one or more example embodiments of the present disclosure.

DETAILED DESCRIPTION

The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, algorithm, and other changes. Portions and features of some embodiments may be included in, or substituted for, those of other embodiments. Embodiments set forth in the claims encompass all available equivalents of those claims.

The Wireless Broadband Alliance's OpenRoaming™ framework is a roaming federal service for providing Wi-Fi® roaming on a global scale through the "One Global Wi-Fi Network." The OpenRoaming™ framework presently supports a cloud-based federation of Access Network Providers (ANPs) and Identity Providers (IdPs). However, a mechanism to provision mobile devices for the OpenRoaming™ framework on a global scale does not presently exist.

The present mechanisms for getting user devices provisioned for the OpenRoaming™ network include various options, such as the Passpoint Online Signup server, captive portals, and other proprietary onboarding solutions at the provider and/or operator level.

For example, the Passpoint Online Signup mechanism has yet to be fully implemented by many device vendors and/or operators due to its complexity. Further, many public Wi-Fi network providers and IdPs are unable to undertake the installation and maintenance of the Passpoint Online Signup mechanism. Further, captive portals tend to be more tedious because they require a multi-step process that includes identifying a network, connecting to the network, launching a web browser, and then filling out access network-specific information.

It would thus be beneficial to replace the present mechanism for onboarding user devices to the OpenRoaming™ framework with a mechanism that allows for quick and secure onboarding of user devices to the OpenRoaming™ framework on a global scale by leveraging the existing public key infrastructure (PKI) trust model that presently supports ANPs and IdPs.

Since there is presently no mechanism for provisioning user devices to the OpenRoaming™ framework on a global scale, the existing PKI trust model that is used to support ANPs and IdPs may be used to onboard user devices. Such a mechanism enables a user device to be ready to establish a connection once it detects a network by automatically using user credentials associated with the user device to identify the network and initiate authentication processes.

Example embodiments of the present disclosure relate to systems, methods, and devices for a mechanism to enable a federated onboarding service in an OpenRoaming™ framework.

In one embodiment, a federated onboarding service may facilitate a mechanism for onboarding a user device to a framework.

In one or more embodiments, a device may receive a prompt to initiate onboarding of the device to a framework. The prompt to initiate onboarding of the device to the framework may be received in response to the device receiving an advertisement associated with the framework

from an ANP. The prompt to initiate onboarding of the device to the framework may be caused by a user associated with the device.

In one or more embodiments, the device may then initiate a federated onboarding service process at an ANP that is connected to the device, where the federated onboarding service process is configured to onboard the device to the framework. The ANP may already be registered with the framework. If the ANP is already registered with the framework, the ANP was previously issued a certificate during the registration process, and the certificate may be used by the device to verify that the ANP is registered with the framework.

In one or more embodiments, the device may then receive a list of available IdPs configured for the framework. The list of available IdPs configured from the framework may include at least one profile associated with a user associated with the device. For example, the list of available IdPs may include an account profile associated with a social media account or an account profile associated with a login for a web portal.

In other embodiments, the device may subsequently receive a selection of an IdP out of the list of available IdPs. The selection of the IdP may be indicated by a user associated with the device.

In one or more embodiments, the device may then establish a connection between the IdP and the device. The establishment of the connection may include a tunnel being established between a federated onboarding service and the device, information being transmitted from the device to the federated onboarding service, and the federated onboarding service being configured to route onboarding of the device to the selected IdP. The information may include at least one of IdP-specific credentials, terms and conditions, or user account validation or creation.

In one or more embodiments, the IdP may generate a user-specific profile associated with the IdP, which may be received from the device.

In one or more embodiments, the device may connect to the framework using the user-specific profile. Prior to the device connecting to the framework, the user-specific profile may be used to authenticate the user.

The proposed solution enables a mechanism to enable mobile devices to be provisioned for the OpenRoaming™ framework on a global scale. Such a mechanism not only provides a uniform onboarding service for users, but also avoids putting the burden on various operators and/or providers to deploy and maintain their own onboarding solutions. This increases overall accessibility to the OpenRoaming™ framework by adding the convenience of using a uniform federated onboarding service and simplifying the onboarding process for users.

The above descriptions are for purposes of illustration and are not meant to be limiting. Numerous other examples, configurations, processes, algorithms, etc., may exist, some of which are described in greater detail below. Example embodiments will now be described with reference to the accompanying figures.

FIG. 1 is a network diagram illustrating an example network environment of roaming among enterprises, according to some example embodiments of the present disclosure. Wireless network 100 may include one or more user devices 120 and one or more access points (AP) 102, which may communicate in accordance with IEEE 802.11 communication standards. The user device(s) 120 may be mobile devices that are non-stationary (e.g., not having fixed locations) or may be stationary devices.

In some embodiments, the user devices 120 and the AP 102 may include one or more computer systems similar to that of the functional diagram of FIG. 6 and/or the example machine/system of FIG. 7.

Referring to FIG. 1, there is shown an Enterprise 1 and an Enterprise 2 that are coordinated through OpenRoaming™ based on the Wireless Broadband Alliance (WBA). Enterprises 1 and 2 are examples of ANPs. In this case, the one or more devices 120 may be able to utilize a federated onboarding service 142 to access networks at either enterprise 1 or 2 based on OpenRoaming provision, authentication, and authorization performed according to FIGS. 2-4. The federated onboarding service 142 may implement OpenRoaming™ functionalities on the behalf of the ANPs and IdPs and enable enterprise to enterprise onboarding and provisioning. For example, a user device 120 may be provisioned for the OpenRoaming™ framework on a global scale through the federated onboarding service 142. Such a mechanism not only provides a uniform onboarding service for users, but also avoids putting the burden on various operators and/or providers to deploy and maintain their own onboarding solutions. This increases overall accessibility to the OpenRoaming™ framework for user device(s) 120 by adding the convenience of using a uniform federated onboarding service and simplifying the onboarding process for users.

One or more illustrative user device(s) 120 and/or AP(s) 102 may be operable by one or more user(s) 110. It should be noted that any addressable unit may be a station (STA). An STA may take on multiple distinct characteristics, each of which shapes its function. For example, a single addressable unit might simultaneously be a portable STA, a quality-of-service (QoS) STA, a dependent STA, and a hidden STA. The one or more illustrative user device(s) 120 and the AP(s) 102 may be STAs. The one or more illustrative user device(s) 120 and/or AP(s) 102 may operate as a personal basic service set (PBSS) control point/access point (PCP/AP). The user device(s) 120 (e.g., 124, 126, or 128) and/or AP(s) 102 may include any suitable processor-driven device including, but not limited to, a mobile device or a non-mobile, e.g., a static device. For example, user device(s) 120 and/or AP(s) 102 may include, a user equipment (UE), a station (STA), an access point (AP), a software enabled AP (SoftAP), a personal computer (PC), a wearable wireless device (e.g., bracelet, watch, glasses, ring, etc.), a desktop computer, a mobile computer, a laptop computer, an Ultra-book™ computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, an internet of things (IoT) device, a sensor device, a PDA device, a handheld PDA device, an on-board device, an off-board device, a hybrid device (e.g., combining cellular phone functionalities with PDA device functionalities), a consumer device, a vehicular device, a non-vehicular device, a mobile or portable device, a non-mobile or non-portable device, a mobile phone, a cellular telephone, a PCS device, a PDA device which incorporates a wireless communication device, a mobile or portable GPS device, a DVB device, a relatively small computing device, a non-desktop computer, a “carry small live large” (CSLL) device, an ultra mobile device (UMD), an ultra mobile PC (UMPC), a mobile internet device (MID), an “origami” device or computing device, a device that supports dynamically composable computing (DCC), a context-aware device, a video device, an audio device, an A/V device, a set-top-box (STB), a blu-ray disc (BD) player, a BD recorder, a digital video disc (DVD) player, a high definition (HD) DVD player, a DVD recorder, a HD DVD recorder, a personal video

recorder (PVR), a broadcast HD receiver, a video source, an audio source, a video sink, an audio sink, a stereo tuner, a broadcast radio receiver, a flat panel display, a personal media player (PMP), a digital video camera (DVC), a digital audio player, a speaker, an audio receiver, an audio amplifier, a gaming device, a data source, a data sink, a digital still camera (DSC), a media player, a smartphone, a television, a music player, or the like. Other devices, including smart devices such as lamps, climate control, car components, household components, appliances, etc. may also be included in this list.

As used herein, the term “Internet of Things (IoT) device” is used to refer to any object (e.g., an appliance, a sensor, etc.) that has an addressable interface (e.g., an Internet protocol (IP) address, a Bluetooth identifier (ID), a near-field communication (NFC) ID, etc.) and can transmit information to one or more other devices over a wired or wireless connection. An IoT device may have a passive communication interface, such as a quick response (QR) code, a radio-frequency identification (RFID) tag, an NFC tag, or the like, or an active communication interface, such as a modem, a transceiver, a transmitter-receiver, or the like. An IoT device can have a particular set of attributes (e.g., a device state or status, such as whether the IoT device is on or off, open or closed, idle or active, available for task execution or busy, and so on, a cooling or heating function, an environmental monitoring or recording function, a light-emitting function, a sound-emitting function, etc.) that can be embedded in and/or controlled/monitored by a central processing unit (CPU), microprocessor, ASIC, or the like, and configured for connection to an IoT network such as a local ad-hoc network or the Internet. For example, IoT devices may include, but are not limited to, refrigerators, toasters, ovens, microwaves, freezers, dishwashers, dishes, hand tools, clothes washers, clothes dryers, furnaces, air conditioners, thermostats, televisions, light fixtures, vacuum cleaners, sprinklers, electricity meters, gas meters, etc., so long as the devices are equipped with an addressable communications interface for communicating with the IoT network. IoT devices may also include cell phones, desktop computers, laptop computers, tablet computers, personal digital assistants (PDAs), etc. Accordingly, the IoT network may be comprised of a combination of “legacy” Internet-accessible devices (e.g., laptop or desktop computers, cell phones, etc.) in addition to devices that do not typically have Internet-connectivity (e.g., dishwashers, etc.).

The user device(s) **120** and/or AP(s) **102** may also include mesh stations in, for example, a mesh network, in accordance with one or more IEEE 802.11 standards and/or 3GPP standards.

Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to communicate with each other via one or more communications networks **130** and/or **135** wirelessly or wired. The user device(s) **120** may also communicate peer-to-peer or directly with each other with or without the AP(s) **102**. Any of the communications networks **130** and/or **135** may include, but not limited to, any one of a combination of different types of suitable communications networks such as, for example, broadcasting networks, cable networks, public networks (e.g., the Internet), private networks, wireless networks, cellular networks, or any other suitable private and/or public networks. Further, any of the communications networks **130** and/or **135** may have any suitable communication range associated therewith and may include, for example, global networks (e.g., the Internet), metropolitan area networks (MANs), wide area networks (WANs), local area networks (LANs), or personal

area networks (PANs). In addition, any of the communications networks **130** and/or **135** may include any type of medium over which network traffic may be carried including, but not limited to, coaxial cable, twisted-pair wire, optical fiber, a hybrid fiber coaxial (HFC) medium, microwave terrestrial transceivers, radio frequency communication mediums, white space communication mediums, ultrahigh frequency communication mediums, satellite communication mediums, or any combination thereof.

Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**) and AP(s) **102** may include one or more communications antennas. The one or more communications antennas may be any suitable type of antennas corresponding to the communications protocols used by the user device(s) **120** (e.g., user devices **124**, **126** and **128**), and AP(s) **102**. Some non-limiting examples of suitable communications antennas include Wi-Fi antennas, Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards compatible antennas, directional antennas, non-directional antennas, dipole antennas, folded dipole antennas, patch antennas, multiple-input multiple-output (MIMO) antennas, omnidirectional antennas, quasi-omnidirectional antennas, or the like. The one or more communications antennas may be communicatively coupled to a radio component to transmit and/or receive signals, such as communications signals to and/or from the user devices **120** and/or AP(s) **102**.

Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform directional transmission and/or directional reception in conjunction with wirelessly communicating in a wireless network. Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform such directional transmission and/or reception using a set of multiple antenna arrays (e.g., DMG antenna arrays or the like). Each of the multiple antenna arrays may be used for transmission and/or reception in a particular respective direction or range of directions. Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform any given directional transmission towards one or more defined transmit sectors. Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform any given directional reception from one or more defined receive sectors.

MIMO beamforming in a wireless network may be accomplished using RF beamforming and/or digital beamforming. In some embodiments, in performing a given MIMO transmission, user devices **120** and/or AP(s) **102** may be configured to use all or a subset of its one or more communications antennas to perform MIMO beamforming.

Any of the user devices **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may include any suitable radio and/or transceiver for transmitting and/or receiving radio frequency (RF) signals in the bandwidth and/or channels corresponding to the communications protocols utilized by any of the user device(s) **120** and AP(s) **102** to communicate with each other. The radio components may include hardware and/or software to modulate and/or demodulate communications signals according to pre-established transmission protocols. The radio components may further have hardware and/or software instructions to communicate via one or more Wi-Fi and/or Wi-Fi direct protocols, as standardized by the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards. In certain example embodiments, the radio component, in cooperation with the communications antennas, may be configured to communicate via 2.4 GHz channels (e.g. 802.11b, 802.11g, 802.11n, 802.11ax), 5 GHz channels (e.g. 802.11n, 802.11ac, 802.11ax), or 60 GHz channels

(e.g. 802.11ad, 802.11ay). 800 MHz channels (e.g. 802.11ah). The communications antennas may operate at 28 GHz and 40 GHz. It should be understood that this list of communication channels in accordance with certain 802.11 standards is only a partial list and that other 802.11 standards may be used (e.g., Next Generation Wi-Fi, or other standards). In some embodiments, non-Wi-Fi protocols may be used for communications between devices, such as Bluetooth, dedicated short-range communication (DSRC), Ultra-High Frequency (UHF) (e.g. IEEE 802.11af, IEEE 802.22), white band frequency (e.g., white spaces), or other packetized radio communications. The radio component may include any known receiver and baseband suitable for communicating via the communications protocols. The radio component may further include a low noise amplifier (LNA), additional signal amplifiers, an analog-to-digital (A/D) converter, one or more buffers, and digital baseband.

It is understood that the above descriptions are for purposes of illustration and are not meant to be limiting.

FIG. 2 depicts an illustrative schematic diagram for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

Current mechanisms are limited because they do not enable mobile devices to be provisioned for the OpenRoaming™ framework on a global scale. Presently, the provisioning of mobile devices continues to rely on proprietary onboarding deployments that vary from one network to another network. For example, the Wi-Fi Alliance has released the Passpoint R2 system, which provides an Online Signup solution to provision mobile devices for public Wi-Fi networks. However, because of the complexity of the Passpoint R2 system, many device vendors have not fully implemented the Passpoint R2 system and many operators have not deployed the Passpoint R2 system. Further, many public Wi-Fi network providers and IdPs are unable to install and maintain the Passpoint R2 system. Thus, the Passpoint R2 system is limited because it enables a localized solution based on the vendors and/or operators who implement the Passpoint R2 system. An alternative solution includes using captive portals to provision mobile devices for the OpenRoaming™ framework. However, captive portals require a different implementation and deployment for each Wi-Fi network. Captive portals are also complex to implement because they require hijacking a session and manual user intervention in identifying a network, connecting to the network, and then launching a web browser and filling out access network-specific information. In the alternative, other proprietary onboarding solutions may be used by each device vendor and/or operator.

As a result, current mechanisms do not provide a reliable, simple, and scalable solution that can support the onboarding of mobile devices for the OpenRoaming™ framework on a global scale.

FIG. 2 depicts an onboarding solution 200 for mobile devices for the OpenRoaming™ framework on a global scale that allows device vendors to implement the onboarding solution 200 and operators to deploy the onboarding solution 200 by leveraging the OpenRoaming™ framework itself. Such a federated onboarding service globally leverages the OpenRoaming™ framework to provision mobile devices by ensuring convenient and uniform implementation for device vendors and increased availability for ANPs 202 and IdPs 204 around the world. Users of mobile devices may therefore enjoy a consistent and simplified user experience when using the federated onboarding service. The onboarding solution 200 will add to the OpenRoaming™ frame-

work's present capabilities to support roaming services for ANPs 202 and IdPs 204 globally.

As depicted in FIG. 2, ANPs 202 may include operators, hospitality and convention centers, airports and other modes of transportation, education centers, cities, governments, retail locations, restaurants, coffee shops, sports stadiums, arenas, corporate offices, venues having public guest Wi-Fi networks, and any other Wi-Fi network provider. IdPs 204 may include mobile operators, cable operators, Internet Service Providers (ISPs), brand-loyalty programs, device-chipset manufacturers, social media providers, public guest Wi-Fi providers, and any other identity provider.

The OpenRoaming™ framework is based on a public key infrastructure (PKI) trust model, where ANPs 202 and IdPs 204 register and onboard with the OpenRoaming™ framework, and registered ANPs 202 and IdPs 204 receive a certificate 206 for use in future verification and authentication processes. The OpenRoaming™ framework may include certificate policy and management capabilities, operated cloud federation capabilities, and policy enabled federation capabilities. The PKI infrastructure that is presently used to support ANP 202 and IdP 204 validation thus enables ANPs 202 and IdPs 204 to enable customers with devices configured for OpenRoaming™ and the appropriate credentials to connect to various Wi-Fi networks through an automatic and secure authentication process. The devices may, for example, be user device(s) 120 as depicted in FIG. 1.

FIG. 3 depicts an illustrative schematic diagram for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

As depicted in FIG. 3, an onboarding solution 300 may include at least one ANP 302, at least one IdP 304, and a federated onboarding service 306. The federated onboarding service 306 may be a web server hosted within the OpenRoaming™ cloud. The federated onboarding service 306 may also hold a certificate signed by the OpenRoaming™ root certificate authority. The federated onboarding service 306 may further be configured to be accessible from anywhere in the world by all participants of the OpenRoaming™ framework, including ANPs 302, IdPs 304, and mobile devices 308 anywhere in the world that are configured for the OpenRoaming™ framework. The mobile devices 308 may include the user device(s) 120 depicted in FIG. 1. Additionally, the federated onboarding service 306 may support a Hypertext Transfer Protocol Secure (HTTPS) protocol. The federated onboarding service 306 may also leverage the present secure OpenRoaming™ PKI infrastructure, which supports validation of ANPs 302 and IdPs 304, to extend the OpenRoaming™ PKI infrastructure for use in onboarding mobile devices 308.

The federated onboarding service 306 may be configured to be capable of communicating with ANPs 302, IdPs 304, and domain name system servers 310. Once each of the ANPs 302 and IdPs 304 are configured for the OpenRoaming™ framework, the federated onboarding service 306 may be used by the ANPs 302 and the IdPs 304, which eliminates the need for each ANP 302 and each IdP 304 to implement and deploy its own onboarding solution.

When a mobile device 308 is being onboarded, the mobile device 308 may validate and/or authenticate the federated onboarding service 306 by using an OpenRoaming™ root certificate installed at the mobile device 308. Following validation and authentication, a secure tunnel between the federated onboarding service 306 and the mobile device 308 may be established using the HTTPS protocol. Following the establishment of the secure tunnel, the root certificate

may no longer be needed. This tunnel between the federated onboarding service 306 and the mobile device 308 differs from tunnels that may be established between ANPs 302 and IdPs 304 within the OpenRoaming™ framework because the tunnels between ANPs 302 and IdPs 304 may use RADSec protocols for authentication purposes.

As depicted in FIG. 3, an OpenRoaming™ framework may already include a DNS server 310 that is used to discover available IdPs 304. While DNS servers 310 are presently used during the authentication process, the DNS servers 310 may be further leveraged during the onboarding process as well. IdPs 304 that are interested in assisting mobile devices 308 during the onboarding process may register with the DNS servers 310 within the OpenRoaming™ framework, thus connecting a user's various identities to the DNS servers 310. Subsequently, when a mobile device 308 is being onboarded, the federated onboarding service 306 may query the DNS server 310 for a list of available IdPs 304 that can support onboarding processing. When an IdP is selected out of the list of available IdPs 304 at the mobile device 308, onboarding processing may be routed to the selected IdP.

Onboarding processing may include account creation for new users, account validation of current users, generation of a user's OpenRoaming™ profile, and/or other applicable functions.

As further depicted in FIG. 3, ANPs 302 may be configured for the OpenRoaming™ framework and may use the federated onboarding service 306. An ANP 302 may opt to register with the OpenRoaming™ framework, and the ANP 302 may be issued a certificate during the registration process. This certificate may be used by mobile devices 308 to validate that the ANP 302 is part of the OpenRoaming™ framework. This certificate may be the same certificate that is used as part of the PKI trust model between ANPs 302 and IdPs 304 during authentication and roaming processing. In some instances, the registration of an ANP 302 and an IdP 304 with the OpenRoaming™ framework may be mandatory, so all ANPs 302 and IdPs 304 may be automatically registered.

Each ANP 302 may configure its network for customized onboarding options. Thus, in one example, an ANP 302 associated with Entity-I may opt to redirect onboarding to only IdPs 304 associated with Entity-I. In another example, an ANP 302 associated with an ISP may support onboarding of mobile devices 308 associated with the ISP's customers and partners. In yet another example, an ANP 302 may not be directly associated with any particular IdP 304, and the ANP 302 may accept all IdPs 304 configured for the OpenRoaming™ framework, thus leaving a user associated with the mobile device 308 to select the IdP 304 that will be used to provision the mobile device 308. As a result, when an onboarding request is received, an ANP 302 may opt to only provide the federated onboarding service 306 such that processing is redirected to an IdP 304 that is associated with an entity associated with the ANP 302. Alternatively, an ANP 302 may opt to provide a list of select partners from which a user associated with the mobile device 308 may select. Further, the ANP 302 may pass the onboarding request along to the federated onboarding service 306 for the federated onboarding service 306 to provide a list of available IdPs 304 that a user associated with the mobile device 308 may select from.

In some instances, an ANP 302 can configure its beacons to indicate that it is part of the OpenRoaming™ framework and is capable of collaborating with the federated onboarding service 306.

Additionally, each mobile device 308 may be issued a certificate from the OpenRoaming™ framework that may be incorporated into the Operating System (OS) of the mobile device 308. This certificate may be used to validate the federated onboarding service 306. If a mobile device 308 is not already onboarded with the OpenRoaming™ framework, the mobile device may not have an existing profile for the OpenRoaming™ framework. In such instances, the mobile device 308 may be able to identify the presence of the OpenRoaming™ network and discover the presence of other onboarding means, such as ANPs 302, in order to begin the onboarding process.

Once an IdP 304 has been selected, an onboarding request may be routed by the federated onboarding service 306 to the IdP 304 where final processing may take place. If a user associated with the mobile device 308 does not presently have an established account with the IdP 304, the user may be prompted to undergo account creation, account validation, profile generation, installation steps, and any other steps necessary for onboarding to be successful. If the user associated with the mobile device 308 presently has an established account with the IdP 304, the user may be validated by the IdP 304, which generates the user's OpenRoaming™ profile and passes the profile to the mobile device 308 through the federated onboarding service 306. In the alternative, a direct secure link may be established between the mobile device 308 and the IdP 304 once the mobile device 308 has validated the federated onboarding service 306 and indicated a selected IdP 304. Once final processing has been completed, the mobile device 308 may be capable of connecting to the OpenRoaming™ framework.

FIG. 4 depicts an illustrative schematic diagram for a federated onboarding service, in accordance with one or more example embodiments of the present disclosure.

As depicted in FIG. 4, an onboarding process 400 may be divided into three main stages. The first stage may be a pre-onboarding stage. A mobile device 408 may have an OpenRoaming™ certificate already installed in the OS of the mobile device 408. However, the mobile device 408 may not have an existing OpenRoaming™ profile at this time. Prior to an onboarding request, an ANP 402 may register with the OpenRoaming™ framework so as to be configured for a federated onboarding service 406. Similarly, an IdP 404 may register with the OpenRoaming™ framework so as to be configured for the federated onboarding service 406. Each ANP 402 and each IdP 404 may have received a certificate during the registration process. The certificate may be unique to each ANP 402 and each IdP 404, but may not be unique to each user that may connect to the ANP 402 and the IdP 404. The IdP 404 may further register with DNS servers 410.

As depicted in FIG. 4, the second stage may be the onboarding stage. In some instances, an ANP 402 that is configured for the federated onboarding service 406 may advertise the OpenRoaming™ framework and/or the federated onboarding service 406 to the mobile device 408. After a user is prompted to onboard the mobile device 408 to the OpenRoaming™ framework, the user may initiate a request for onboarding to the federated onboarding service 406. The mobile device 408 thus may trigger a federated onboarding service process at the ANP 402, which may communicate with the federated onboarding service 406 to undergo the federated onboarding service process. In some instances, if the mobile device 408 is already configured with an Internet connection, the mobile device 408 may directly connect with the federated onboarding service 406 instead of indi-

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rectly connecting with the federated onboarding service **406** through the ANP **402**. In one exemplary method, the federated onboarding service **406** may transmit a list of available IdPs **404** through the ANP **402** to the mobile device **408**. A user associated with the mobile device **408** may select an IdP **404** for the onboarding process, and the selected IdP **404** is then communicated to the federated onboarding service **406**. The federated onboarding service **406** may then communicate with the DNS servers **410** to look up the selected IdP **404**. Subsequently, the mobile device **408** may validate the federated onboarding service **406** before providing information to the federated onboarding service **406**. Information may include user interactions having IdP credentials, account validation and/or creation, terms and conditions that could be specific to IdP **404** or a default OpenRoaming™ terms and conditions provided by the OpenRoaming™ federation, or other applicable information. The federated onboarding service **406** may route the onboarding process to the selected IdP **404** if the IdP **404** is willing to onboard the mobile device **408**. Finally, the IdP **404** may then communicate with the mobile device **408** in order to provision the mobile device **408** for onboarding and the IdP **404** may then create a user-specific profile associated with the OpenRoaming™ framework. The user-specific profile may be stored at the mobile device **408** and may be subsequently used when the user next attempts to access the OpenRoaming™ framework.

As depicted in FIG. 4, the third stage may be the connection stage. During the connection stage, the mobile device **408** may authenticate the ANP **402** as part of the OpenRoaming™ framework. The user-specific profile may be used for the authentication process. Once authentication is complete, the ANP **402** may use the DNS server **410** to search for available IdPs **404**. When an available IdP **404** has been identified, a tunnel may be established between the ANP **402** and the IdP **404** in order to undergo extensible authentication protocol (EAP) authentication using RADSec protocols. Following authentication, access may be granted at the mobile device **408** for the mobile device **408** to connect to the OpenRoaming™ framework. In some instances, once the user-specific profile has been stored at the mobile device **408**, the credentials associated with the user-specific profile may be automatically used to identify a network and initiate authentication in order to connect to the network.

FIG. 5 illustrates a flow diagram of the illustrative process for an illustrative federated onboarding service system, in accordance with one or more example embodiments of the present disclosure.

At block **502**, a device (e.g., the user device(s) **120** FIG. 1) may receive a prompt to initiate onboarding of the device to a framework. The prompt to initiate onboarding of the device to the framework may be received in response to the device receiving an advertisement associated with the framework from an access network provider (ANP).

At block **504**, the device may initiate a federated onboarding service process at the ANP that is connected to the device, where the federated onboarding service process is configured to onboard the device to the framework. The ANP may be registered with the framework. If the ANP is already registered with the framework, the ANP may have been previously issued a certificate during registration, and the certificate may be used by the device to verify that the ANP is registered with the framework.

At block **506**, the device may receive a list of available identity providers (IdPs) configured for the framework. The

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list of available IdPs configured for the framework may include at least one profile associated with a user associated with the device.

At block **508**, the device may receive a selection of an IdP of the list of available IdPs.

At block **510**, the device may establish a connection between the IdP and the device. The establishment of the connection between the IdP and the device may involve establishing a tunnel between a federated onboarding service and the device and then transmitting information from the device to the federated onboarding service. The information may include IdP-specific credentials, terms and conditions, or user account validation or creation. The federated onboarding service may then be configured to route onboarding of the device to the IdP.

At block **512**, a user-specific profile associated with the IdP may be generated by the IdP and received by the device.

At block **514**, the device may connect to the framework using the user-specific profile. The user-specific profile may be used to authenticate the device prior to the device connecting to the framework.

It is understood that the above descriptions are for purposes of illustration and are not meant to be limiting.

FIG. 6 shows a functional diagram of an exemplary communication station **600**, in accordance with one or more example embodiments of the present disclosure. In one embodiment, FIG. 6 illustrates a functional block diagram of a communication station that may be suitable for use as an AP **102** (FIG. 1) or a user device **120** (FIG. 1) in accordance with some embodiments. The communication station **600** may also be suitable for use as a handheld device, a mobile device, a cellular telephone, a smartphone, a tablet, a netbook, a wireless terminal, a laptop computer, a wearable computer device, a femtocell, a high data rate (HDR) subscriber station, an access point, an access terminal, or other personal communication system (PCS) device.

The communication station **600** may include communications circuitry **602** and a transceiver **610** for transmitting and receiving signals to and from other communication stations using one or more antennas **601**. The communications circuitry **602** may include circuitry that can operate the physical layer (PHY) communications and/or medium access control (MAC) communications for controlling access to the wireless medium, and/or any other communications layers for transmitting and receiving signals. The communication station **600** may also include processing circuitry **606** and memory **608** arranged to perform the operations described herein. In some embodiments, the communications circuitry **602** and the processing circuitry **606** may be configured to perform operations detailed in the above figures, diagrams, and flows.

In accordance with some embodiments, the communications circuitry **602** may be arranged to contend for a wireless medium and configure frames or packets for communicating over the wireless medium. The communications circuitry **602** may be arranged to transmit and receive signals. The communications circuitry **602** may also include circuitry for modulation/demodulation, upconversion/downconversion, filtering, amplification, etc. In some embodiments, the processing circuitry **606** of the communication station **600** may include one or more processors. In other embodiments, two or more antennas **601** may be coupled to the communications circuitry **602** arranged for sending and receiving signals. The memory **608** may store information for configuring the processing circuitry **606** to perform operations for configuring and transmitting message frames and performing the various operations described herein. The memory **608**

may include any type of memory, including non-transitory memory, for storing information in a form readable by a machine (e.g., a computer). For example, the memory **608** may include a computer-readable storage device, read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices and other storage devices and media.

In some embodiments, the communication station **600** may be part of a portable wireless communication device, such as a personal digital assistant (PDA), a laptop or portable computer with wireless communication capability, a web tablet, a wireless telephone, a smartphone, a wireless headset, a pager, an instant messaging device, a digital camera, an access point, a television, a medical device (e.g., a heart rate monitor, a blood pressure monitor, etc.), a wearable computer device, or another device that may receive and/or transmit information wirelessly.

In some embodiments, the communication station **600** may include one or more antennas **601**. The antennas **601** may include one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas, or other types of antennas suitable for transmission of RF signals. In some embodiments, instead of two or more antennas, a single antenna with multiple apertures may be used. In these embodiments, each aperture may be considered a separate antenna. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated for spatial diversity and the different channel characteristics that may result between each of the antennas and the antennas of a transmitting station.

In some embodiments, the communication station **600** may include one or more of a keyboard, a display, a non-volatile memory port, multiple antennas, a graphics processor, an application processor, speakers, and other mobile device elements. The display may be an LCD screen including a touch screen.

Although the communication station **600** is illustrated as having several separate functional elements, two or more of the functional elements may be combined and may be implemented by combinations of software-configured elements, such as processing elements including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may include one or more microprocessors, DSPs, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs) and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements of the communication station **600** may refer to one or more processes operating on one or more processing elements.

Certain embodiments may be implemented in one or a combination of hardware, firmware, and software. Other embodiments may also be implemented as instructions stored on a computer-readable storage device, which may be read and executed by at least one processor to perform the operations described herein. A computer-readable storage device may include any non-transitory memory mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a computer-readable storage device may include read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices, and other storage devices and media. In some embodiments, the communica-

tion station **600** may include one or more processors and may be configured with instructions stored on a computer-readable storage device.

FIG. 7 illustrates a block diagram of an example of a machine **700** or system upon which any one or more of the techniques (e.g., methodologies) discussed herein may be performed. In other embodiments, the machine **700** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine **700** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **700** may act as a peer machine in peer-to-peer (P2P) (or other distributed) network environments. The machine **700** may be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile telephone, a wearable computer device, a web appliance, a network router, a switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine, such as a base station. Further, while only a single machine is illustrated, the term "machine" shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), or other computer cluster configurations.

Examples, as described herein, may include or may operate on logic or a number of components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations when operating. A module includes hardware. In an example, the hardware may be specifically configured to carry out a specific operation (e.g., hardwired). In another example, the hardware may include configurable execution units (e.g., transistors, circuits, etc.) and a computer readable medium containing instructions where the instructions configure the execution units to carry out a specific operation when in operation. The configuring may occur under the direction of the executions units or a loading mechanism. Accordingly, the execution units are communicatively coupled to the computer-readable medium when the device is operating. In this example, the execution units may be a member of more than one module. For example, under operation, the execution units may be configured by a first set of instructions to implement a first module at one point in time and reconfigured by a second set of instructions to implement a second module at a second point in time.

The machine (e.g., computer system) **700** may include a hardware processor **702** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **704** and a static memory **706**, some or all of which may communicate with each other via an interlink (e.g., bus) **708**. The machine **700** may further include a power management device **732**, a graphics display device **710**, an alphanumeric input device **712** (e.g., a keyboard), and a user interface (UI) navigation device **714** (e.g., a mouse). In an example, the graphics display device **710**, alphanumeric input device **712**, and UI navigation device **714** may be a touch screen display. The machine **700** may additionally include a storage device (i.e., drive unit) **716**, a signal generation device **718** (e.g., a speaker), a federated onboarding service device **719**, a network interface device/transceiver **720** coupled to antenna(s) **730**, and one or more sensors **728**, such as a global positioning system (GPS) sensor, a compass, an accelerometer, or other sensor. The machine **700** may include an output controller **734**, such as a serial (e.g.,

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universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate with or control one or more peripheral devices (e.g., a printer, a card reader, etc.)). The operations in accordance with one or more example embodiments of the present disclosure may be carried out by a baseband processor. The baseband processor may be configured to generate corresponding baseband signals. The baseband processor may further include physical layer (PHY) and medium access control layer (MAC) circuitry, and may further interface with the hardware processor **702** for generation and processing of the baseband signals and for controlling operations of the main memory **704**, the storage device **716**, and/or the federated onboarding service device **719**. The baseband processor may be provided on a single radio card, a single chip, or an integrated circuit (IC).

The storage device **716** may include a machine readable medium **722** on which is stored one or more sets of data structures or instructions **724** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **724** may also reside, completely or at least partially, within the main memory **704**, within the static memory **706**, or within the hardware processor **702** during execution thereof by the machine **700**. In an example, one or any combination of the hardware processor **702**, the main memory **704**, the static memory **706**, or the storage device **716** may constitute machine-readable media.

The federated onboarding service device **719** may carry out or perform any of the operations and processes (e.g., process **500**) described and shown above.

It is understood that the above are only a subset of what the federated onboarding service device **719** may be configured to perform and that other functions included throughout this disclosure may also be performed by the federated onboarding service device **719**.

While the machine-readable medium **722** is illustrated as a single medium, the term “machine-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **724**.

Various embodiments may be implemented fully or partially in software and/or firmware. This software and/or firmware may take the form of instructions contained in or on a non-transitory computer-readable storage medium. Those instructions may then be read and executed by one or more processors to enable performance of the operations described herein. The instructions may be in any suitable form, such as but not limited to source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. Such a computer-readable medium may include any tangible non-transitory medium for storing information in a form readable by one or more computers, such as but not limited to read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; a flash memory, etc.

The term “machine-readable medium” may include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **700** and that cause the machine **700** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding, or carrying data structures used by or associated with such instructions. Non-limiting machine-readable medium examples may include solid-state memories and optical and magnetic media. In an example, a massed machine-readable medium includes a machine-readable

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medium with a plurality of particles having resting mass. Specific examples of massed machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., electrically programmable read-only memory (EPROM), or electrically erasable programmable read-only memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks.

The instructions **724** may further be transmitted or received over a communications network **726** using a transmission medium via the network interface device/transceiver **720** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communications networks may include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), plain old telephone (POTS) networks, wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.16 family of standards known as WiMax®, IEEE 802.15.4 family of standards, and peer-to-peer (P2P) networks, among others. In an example, the network interface device/transceiver **720** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **726**. In an example, the network interface device/transceiver **720** may include a plurality of antennas to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine **700** and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

The operations and processes described and shown above may be carried out or performed in any suitable order as desired in various implementations. Additionally, in certain implementations, at least a portion of the operations may be carried out in parallel. Furthermore, in certain implementations, less than or more than the operations described may be performed.

FIG. **8** is a block diagram of a radio architecture **105A**, **105B** in accordance with some embodiments that may be implemented in any one of the example APs **102** and/or the example user device(s) **120** of FIG. **1**. Radio architecture **105A**, **105B** may include radio front-end module (FEM) circuitry **804a-b**, radio IC circuitry **806a-b** and baseband processing circuitry **808a-b**. Radio architecture **105A**, **105B** as shown includes both Wireless Local Area Network (WLAN) functionality and Bluetooth (BT) functionality although embodiments are not so limited. In this disclosure, “WLAN” and “Wi-Fi” are used interchangeably.

FEM circuitry **804a-b** may include a WLAN or Wi-Fi FEM circuitry **804a** and a Bluetooth (BT) FEM circuitry **804b**. The WLAN FEM circuitry **804a** may include a receive signal path comprising circuitry configured to operate on WLAN RF signals received from one or more antennas **801**, to amplify the received signals and to provide the amplified versions of the received signals to the WLAN radio IC circuitry **806a** for further processing. The BT FEM circuitry **804b** may include a receive signal path which may include circuitry configured to operate on BT RF signals received

from one or more antennas **801**, to amplify the received signals and to provide the amplified versions of the received signals to the BT radio IC circuitry **806b** for further processing. FEM circuitry **804a** may also include a transmit signal path which may include circuitry configured to amplify WLAN signals provided by the radio IC circuitry **806a** for wireless transmission by one or more of the antennas **801**. In addition, FEM circuitry **804b** may also include a transmit signal path which may include circuitry configured to amplify BT signals provided by the radio IC circuitry **806b** for wireless transmission by the one or more antennas. In the embodiment of FIG. 8, although FEM **804a** and FEM **804b** are shown as being distinct from one another, embodiments are not so limited, and include within their scope the use of an FEM (not shown) that includes a transmit path and/or a receive path for both WLAN and BT signals, or the use of one or more FEM circuitries where at least some of the FEM circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

Radio IC circuitry **806a-b** as shown may include WLAN radio IC circuitry **806a** and BT radio IC circuitry **806b**. The WLAN radio IC circuitry **806a** may include a receive signal path which may include circuitry to down-convert WLAN RF signals received from the FEM circuitry **804a** and provide baseband signals to WLAN baseband processing circuitry **808a**. BT radio IC circuitry **806b** may in turn include a receive signal path which may include circuitry to down-convert BT RF signals received from the FEM circuitry **804b** and provide baseband signals to BT baseband processing circuitry **808b**. WLAN radio IC circuitry **806a** may also include a transmit signal path which may include circuitry to up-convert WLAN baseband signals provided by the WLAN baseband processing circuitry **808a** and provide WLAN RF output signals to the FEM circuitry **804a** for subsequent wireless transmission by the one or more antennas **801**. BT radio IC circuitry **806b** may also include a transmit signal path which may include circuitry to up-convert BT baseband signals provided by the BT baseband processing circuitry **808b** and provide BT RF output signals to the FEM circuitry **804b** for subsequent wireless transmission by the one or more antennas **801**. In the embodiment of FIG. 8, although radio IC circuitries **806a** and **806b** are shown as being distinct from one another, embodiments are not so limited, and include within their scope the use of a radio IC circuitry (not shown) that includes a transmit signal path and/or a receive signal path for both WLAN and BT signals, or the use of one or more radio IC circuitries where at least some of the radio IC circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

Baseband processing circuitry **808a-b** may include a WLAN baseband processing circuitry **808a** and a BT baseband processing circuitry **808b**. The WLAN baseband processing circuitry **808a** may include a memory, such as, for example, a set of RAM arrays in a Fast Fourier Transform or Inverse Fast Fourier Transform block (not shown) of the WLAN baseband processing circuitry **808a**. Each of the WLAN baseband circuitry **808a** and the BT baseband circuitry **808b** may further include one or more processors and control logic to process the signals received from the corresponding WLAN or BT receive signal path of the radio IC circuitry **806a-b**, and to also generate corresponding WLAN or BT baseband signals for the transmit signal path of the radio IC circuitry **806a-b**. Each of the baseband processing circuitries **808a** and **808b** may further include physical layer (PHY) and medium access control layer (MAC) circuitry, and may further interface with a device for generation and

processing of the baseband signals and for controlling operations of the radio IC circuitry **806a-b**.

Referring still to FIG. 8, according to the shown embodiment, WLAN-BT coexistence circuitry **813** may include logic providing an interface between the WLAN baseband circuitry **808a** and the BT baseband circuitry **808b** to enable use cases requiring WLAN and BT coexistence. In addition, a switch **803** may be provided between the WLAN FEM circuitry **804a** and the BT FEM circuitry **804b** to allow switching between the WLAN and BT radios according to application needs. In addition, although the antennas **801** are depicted as being respectively connected to the WLAN FEM circuitry **804a** and the BT FEM circuitry **804b**, embodiments include within their scope the sharing of one or more antennas as between the WLAN and BT FEMs, or the provision of more than one antenna connected to each of FEM **804a** or **804b**.

In some embodiments, the front-end module circuitry **804a-b**, the radio IC circuitry **806a-b**, and baseband processing circuitry **808a-b** may be provided on a single radio card, such as wireless radio card **802**. In some other embodiments, the one or more antennas **801**, the FEM circuitry **804a-b** and the radio IC circuitry **806a-b** may be provided on a single radio card. In some other embodiments, the radio IC circuitry **806a-b** and the baseband processing circuitry **808a-b** may be provided on a single chip or integrated circuit (IC), such as IC **812**.

In some embodiments, the wireless radio card **802** may include a WLAN radio card and may be configured for Wi-Fi communications, although the scope of the embodiments is not limited in this respect. In some of these embodiments, the radio architecture **105A**, **105B** may be configured to receive and transmit orthogonal frequency division multiplexed (OFDM) or orthogonal frequency division multiple access (OFDMA) communication signals over a multicarrier communication channel. The OFDM or OFDMA signals may comprise a plurality of orthogonal subcarriers.

In some of these multicarrier embodiments, radio architecture **105A**, **105B** may be part of a Wi-Fi communication station (STA) such as a wireless access point (AP), a base station or a mobile device including a Wi-Fi device. In some of these embodiments, radio architecture **105A**, **105B** may be configured to transmit and receive signals in accordance with specific communication standards and/or protocols, such as any of the Institute of Electrical and Electronics Engineers (IEEE) standards including, 802.11n-2009, IEEE 802.11-2012, IEEE 802.11-2016, 802.11n-2009, 802.11ac, 802.11ah, 802.11ad, 802.11ay and/or 802.11ax standards and/or proposed specifications for WLANs, although the scope of embodiments is not limited in this respect. Radio architecture **105A**, **105B** may also be suitable to transmit and/or receive communications in accordance with other techniques and standards.

In some embodiments, the radio architecture **105A**, **105B** may be configured for high-efficiency Wi-Fi (HEW) communications in accordance with the IEEE 802.11ax standard. In these embodiments, the radio architecture **105A**, **105B** may be configured to communicate in accordance with an OFDMA technique, although the scope of the embodiments is not limited in this respect.

In some other embodiments, the radio architecture **105A**, **105B** may be configured to transmit and receive signals transmitted using one or more other modulation techniques such as spread spectrum modulation (e.g., direct sequence code division multiple access (DS-CDMA) and/or frequency hopping code division multiple access (FH-CDMA),

time-division multiplexing (TDM) modulation, and/or frequency-division multiplexing (FDM) modulation, although the scope of the embodiments is not limited in this respect.

In some embodiments, as further shown in FIG. 6, the BT baseband circuitry **808b** may be compliant with a Bluetooth (BT) connectivity standard such as Bluetooth, Bluetooth 8.0 or Bluetooth 6.0, or any other iteration of the Bluetooth Standard.

In some embodiments, the radio architecture **105A**, **105B** may include other radio cards, such as a cellular radio card configured for cellular (e.g., 5GPP such as LTE, LTE-Advanced or 7G communications).

In some IEEE 802.11 embodiments, the radio architecture **105A**, **105B** may be configured for communication over various channel bandwidths including bandwidths having center frequencies of about 900 MHz, 2.4 GHz, 5 GHz, and bandwidths of about 2 MHz, 4 MHz, 5 MHz, 5.5 MHz, 6 MHz, 8 MHz, 10 MHz, 20 MHz, 40 MHz, 80 MHz (with contiguous bandwidths) or 80+80 MHz (160 MHz) (with non-contiguous bandwidths). In some embodiments, a 920 MHz channel bandwidth may be used. The scope of the embodiments is not limited with respect to the above center frequencies however.

FIG. 9 illustrates WLAN FEM circuitry **804a** in accordance with some embodiments. Although the example of FIG. 9 is described in conjunction with the WLAN FEM circuitry **804a**, the example of FIG. 9 may be described in conjunction with the example BT FEM circuitry **804b** (FIG. 8), although other circuitry configurations may also be suitable.

In some embodiments, the FEM circuitry **804a** may include a TX/RX switch **902** to switch between transmit mode and receive mode operation. The FEM circuitry **804a** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **804a** may include a low-noise amplifier (LNA) **906** to amplify received RF signals **903** and provide the amplified received RF signals **907** as an output (e.g., to the radio IC circuitry **806a-b** (FIG. 8)). The transmit signal path of the circuitry **804a** may include a power amplifier (PA) to amplify input RF signals **909** (e.g., provided by the radio IC circuitry **806a-b**), and one or more filters **912**, such as band-pass filters (BPFs), low-pass filters (LPFs) or other types of filters, to generate RF signals **915** for subsequent transmission (e.g., by one or more of the antennas **801** (FIG. 8) via an example duplexer **914**).

In some dual-mode embodiments for Wi-Fi communication, the FEM circuitry **804a** may be configured to operate in either the 2.4 GHz frequency spectrum or the 5 GHz frequency spectrum. In these embodiments, the receive signal path of the FEM circuitry **804a** may include a receive signal path duplexer **904** to separate the signals from each spectrum as well as provide a separate LNA **906** for each spectrum as shown. In these embodiments, the transmit signal path of the FEM circuitry **804a** may also include a power amplifier **910** and a filter **912**, such as a BPF, an LPF or another type of filter for each frequency spectrum and a transmit signal path duplexer **904** to provide the signals of one of the different spectrums onto a single transmit path for subsequent transmission by the one or more of the antennas **801** (FIG. 8). In some embodiments, BT communications may utilize the 2.4 GHz signal paths and may utilize the same FEM circuitry **804a** as the one used for WLAN communications.

FIG. 10 illustrates radio IC circuitry **806a** in accordance with some embodiments. The radio IC circuitry **806a** is one example of circuitry that may be suitable for use as the

WLAN or BT radio IC circuitry **806a/806b** (FIG. 8), although other circuitry configurations may also be suitable. Alternatively, the example of FIG. 10 may be described in conjunction with the example BT radio IC circuitry **806b**.

In some embodiments, the radio IC circuitry **806a** may include a receive signal path and a transmit signal path. The receive signal path of the radio IC circuitry **806a** may include at least mixer circuitry **1002**, such as, for example, down-conversion mixer circuitry, amplifier circuitry **1006** and filter circuitry **1008**. The transmit signal path of the radio IC circuitry **806a** may include at least filter circuitry **1012** and mixer circuitry **1014**, such as, for example, up-conversion mixer circuitry. Radio IC circuitry **806a** may also include synthesizer circuitry **1004** for synthesizing a frequency **1005** for use by the mixer circuitry **1002** and the mixer circuitry **1014**. The mixer circuitry **1002** and/or **1014** may each, according to some embodiments, be configured to provide direct conversion functionality. The latter type of circuitry presents a much simpler architecture as compared with standard super-heterodyne mixer circuitries, and any flicker noise brought about by the same may be alleviated for example through the use of OFDM modulation. FIG. 10 illustrates only a simplified version of a radio IC circuitry, and may include, although not shown, embodiments where each of the depicted circuitries may include more than one component. For instance, mixer circuitry **1014** may each include one or more mixers, and filter circuitries **1008** and/or **1012** may each include one or more filters, such as one or more BPFs and/or LPFs according to application needs. For example, when mixer circuitries are of the direct-conversion type, they may each include two or more mixers.

In some embodiments, mixer circuitry **1002** may be configured to down-convert RF signals **907** received from the FEM circuitry **804a-b** (FIG. 8) based on the synthesized frequency **1005** provided by synthesizer circuitry **1004**. The amplifier circuitry **1006** may be configured to amplify the down-converted signals and the filter circuitry **1008** may include an LPF configured to remove unwanted signals from the down-converted signals to generate output baseband signals **1007**. Output baseband signals **1007** may be provided to the baseband processing circuitry **808a-b** (FIG. 8) for further processing. In some embodiments, the output baseband signals **1007** may be zero-frequency baseband signals, although this is not a requirement. In some embodiments, mixer circuitry **1002** may comprise passive mixers, although the scope of the embodiments is not limited in this respect.

In some embodiments, the mixer circuitry **1014** may be configured to up-convert input baseband signals **1011** based on the synthesized frequency **1005** provided by the synthesizer circuitry **1004** to generate RF output signals **909** for the FEM circuitry **804a-b**. The baseband signals **1011** may be provided by the baseband processing circuitry **808a-b** and may be filtered by filter circuitry **1012**. The filter circuitry **1012** may include an LPF or a BPF, although the scope of the embodiments is not limited in this respect.

In some embodiments, the mixer circuitry **1002** and the mixer circuitry **1014** may each include two or more mixers and may be arranged for quadrature down-conversion and/or up-conversion respectively with the help of synthesizer **1004**. In some embodiments, the mixer circuitry **1002** and the mixer circuitry **1014** may each include two or more mixers each configured for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **1002** and the mixer circuitry **1014** may be arranged for direct down-conversion and/or direct up-conversion, respectively. In some embodiments, the mixer circuitry **1002** and the

mixer circuitry **1014** may be configured for super-heterodyne operation, although this is not a requirement.

Mixer circuitry **1002** may comprise, according to one embodiment: quadrature passive mixers (e.g., for the in-phase (I) and quadrature phase (Q) paths). In such an embodiment, RF input signal **907** from FIG. 9 may be down-converted to provide I and Q baseband output signals to be sent to the baseband processor.

Quadrature passive mixers may be driven by zero and ninety-degree time-varying LO switching signals provided by a quadrature circuitry which may be configured to receive a LO frequency (fLO) from a local oscillator or a synthesizer, such as LO frequency **1005** of synthesizer **1004** (FIG. 10). In some embodiments, the LO frequency may be the carrier frequency, while in other embodiments, the LO frequency may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the zero and ninety-degree time-varying switching signals may be generated by the synthesizer, although the scope of the embodiments is not limited in this respect.

In some embodiments, the LO signals may differ in duty cycle (the percentage of one period in which the LO signal is high) and/or offset (the difference between start points of the period). In some embodiments, the LO signals may have an 85% duty cycle and an 80% offset. In some embodiments, each branch of the mixer circuitry (e.g., the in-phase (I) and quadrature phase (Q) path) may operate at an 80% duty cycle, which may result in a significant reduction in power consumption.

The RF input signal **907** (FIG. 9) may comprise a balanced signal, although the scope of the embodiments is not limited in this respect. The I and Q baseband output signals may be provided to low-noise amplifier, such as amplifier circuitry **1006** (FIG. 10) or to filter circuitry **1008** (FIG. 10).

In some embodiments, the output baseband signals **1007** and the input baseband signals **1011** may be analog baseband signals, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband signals **1007** and the input baseband signals **1011** may be digital baseband signals. In these alternate embodiments, the radio IC circuitry may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry.

In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing signals for each spectrum, or for other spectrums not mentioned here, although the scope of the embodiments is not limited in this respect.

In some embodiments, the synthesizer circuitry **1004** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. For example, synthesizer circuitry **1004** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider. According to some embodiments, the synthesizer circuitry **1004** may include digital synthesizer circuitry. An advantage of using a digital synthesizer circuitry is that, although it may still include some analog components, its footprint may be scaled down much more than the footprint of an analog synthesizer circuitry. In some embodiments, frequency input into synthesizer circuitry **1004** may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. A divider control input may further be provided by either the baseband processing circuitry **808a-b** (FIG. 8) depending on the desired output frequency

1005. In some embodiments, a divider control input (e.g., N) may be determined from a look-up table (e.g., within a Wi-Fi card) based on a channel number and a channel center frequency as determined or indicated by the example application processor **810**. The application processor **810** may include, or otherwise be connected to, one of the example secure signal converter **101** or the example received signal converter **103** (e.g., depending on which device the example radio architecture is implemented in).

In some embodiments, synthesizer circuitry **1004** may be configured to generate a carrier frequency as the output frequency **1005**, while in other embodiments, the output frequency **1005** may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the output frequency **1005** may be a LO frequency (fLO).

FIG. 11 illustrates a functional block diagram of baseband processing circuitry **808a** in accordance with some embodiments. The baseband processing circuitry **808a** is one example of circuitry that may be suitable for use as the baseband processing circuitry **808a** (FIG. 8), although other circuitry configurations may also be suitable. Alternatively, the example of FIG. 10 may be used to implement the example BT baseband processing circuitry **808b** of FIG. 8.

The baseband processing circuitry **808a** may include a receive baseband processor (RX BBP) **1102** for processing receive baseband signals **1009** provided by the radio IC circuitry **806a-b** (FIG. 8) and a transmit baseband processor (TX BBP) **1104** for generating transmit baseband signals **1011** for the radio IC circuitry **806a-b**. The baseband processing circuitry **808a** may also include control logic **1106** for coordinating the operations of the baseband processing circuitry **808a**.

In some embodiments (e.g., when analog baseband signals are exchanged between the baseband processing circuitry **808a-b** and the radio IC circuitry **806a-b**), the baseband processing circuitry **808a** may include ADC **1110** to convert analog baseband signals **1109** received from the radio IC circuitry **806a-b** to digital baseband signals for processing by the RX BBP **1102**. In these embodiments, the baseband processing circuitry **808a** may also include DAC **1112** to convert digital baseband signals from the TX BBP **1104** to analog baseband signals **1111**.

In some embodiments that communicate OFDM signals or OFDMA signals, such as through baseband processor **808a**, the transmit baseband processor **1104** may be configured to generate OFDM or OFDMA signals as appropriate for transmission by performing an inverse fast Fourier transform (IFFT). The receive baseband processor **1102** may be configured to process received OFDM signals or OFDMA signals by performing an FFT. In some embodiments, the receive baseband processor **1102** may be configured to detect the presence of an OFDM signal or OFDMA signal by performing an autocorrelation, to detect a preamble, such as a short preamble, and by performing a cross-correlation, to detect a long preamble. The preambles may be part of a predetermined frame structure for Wi-Fi communication.

Referring back to FIG. 8, in some embodiments, the antennas **801** (FIG. 8) may each comprise one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas or other types of antennas suitable for transmission of RF signals. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated to take advantage of spatial diversity and the different channel characteristics that

may result. Antennas **801** may each include a set of phased-array antennas, although embodiments are not so limited.

Although the radio architecture **105A**, **105B** is illustrated as having several separate functional elements, one or more of the functional elements may be combined and may be implemented by combinations of software-configured elements, such as processing elements including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may comprise one or more microprocessors, DSPs, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs) and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements may refer to one or more processes operating on one or more processing elements.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. The terms “computing device,” “user device,” “communication station,” “station,” “handheld device,” “mobile device,” “wireless device” and “user equipment” (UE) as used herein refers to a wireless communication device such as a cellular telephone, a smartphone, a tablet, a netbook, a wireless terminal, a laptop computer, a femtocell, a high data rate (HDL) subscriber station, an access point, a printer, a point of sale device, an access terminal, or other personal communication system (PCS) device. The device may be either mobile or stationary.

As used within this document, the term “communicate” is intended to include transmitting, or receiving, or both transmitting and receiving. This may be particularly useful in claims when describing the organization of data that is being transmitted by one device and received by another, but only the functionality of one of those devices is required to infringe the claim. Similarly, the bidirectional exchange of data between two devices (both devices transmit and receive during the exchange) may be described as “communicating,” when only the functionality of one of those devices is being claimed. The term “communicating” as used herein with respect to a wireless communication signal includes transmitting the wireless communication signal and/or receiving the wireless communication signal. For example, a wireless communication unit, which is capable of communicating a wireless communication signal, may include a wireless transmitter to transmit the wireless communication signal to at least one other wireless communication unit, and/or a wireless communication receiver to receive the wireless communication signal from at least one other wireless communication unit.

As used herein, unless otherwise specified, the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicates that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

The term “access point” (AP) as used herein may be a fixed station. An access point may also be referred to as an access node, a base station, an evolved node B (eNodeB), or some other similar terminology known in the art. An access terminal may also be called a mobile station, user equipment (UE), a wireless communication device, or some other similar terminology known in the art. Embodiments disclosed herein generally pertain to wireless networks. Some

embodiments may relate to wireless networks that operate in accordance with one of the IEEE 802.11 standards.

Some embodiments may be used in conjunction with various devices and systems, for example, a personal computer (PC), a desktop computer, a mobile computer, a laptop computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, a personal digital assistant (PDA) device, a handheld PDA device, an on-board device, an off-board device, a hybrid device, a vehicular device, a non-vehicular device, a mobile or portable device, a consumer device, a non-mobile or non-portable device, a wireless communication station, a wireless communication device, a wireless access point (AP), a wired or wireless router, a wired or wireless modem, a video device, an audio device, an audio-video (A/V) device, a wired or wireless network, a wireless area network, a wireless video area network (WVAN), a local area network (LAN), a wireless LAN (WLAN), a personal area network (PAN), a wireless PAN (WPAN), and the like.

Some embodiments may be used in conjunction with one way and/or two-way radio communication systems, cellular radio-telephone communication systems, a mobile phone, a cellular telephone, a wireless telephone, a personal communication system (PCS) device, a PDA device which incorporates a wireless communication device, a mobile or portable global positioning system (GPS) device, a device which incorporates a GPS receiver or transceiver or chip, a device which incorporates an RFID element or chip, a multiple input multiple output (MIMO) transceiver or device, a single input multiple output (SIMO) transceiver or device, a multiple input single output (MISO) transceiver or device, a device having one or more internal antennas and/or external antennas, digital video broadcast (DVB) devices or systems, multi-standard radio devices or systems, a wired or wireless handheld device, e.g., a smartphone, a wireless application protocol (WAP) device, or the like.

Some embodiments may be used in conjunction with one or more types of wireless communication signals and/or systems following one or more wireless communication protocols, for example, radio frequency (RF), infrared (IR), frequency-division multiplexing (FDM), orthogonal FDM (OFDM), time-division multiplexing (TDM), time-division multiple access (TDMA), extended TDMA (E-TDMA), general packet radio service (GPRS), extended GPRS, code-division multiple access (CDMA), wideband CDMA (WCDMA), CDMA 2000, single-carrier CDMA, multi-carrier CDMA, multi-carrier modulation (MDM), discrete multi-tone (DMT), Bluetooth®, global positioning system (GPS), Wi-Fi, Wi-Max, ZigBee, ultra-wideband (UWB), global system for mobile communications (GSM), 2G, 2.5G, 3G, 3.5G, 4G, fifth generation (5G) mobile networks, 3GPP, long term evolution (LTE), LTE advanced, enhanced data rates for GSM Evolution (EDGE), or the like. Other embodiments may be used in various other devices, systems, and/or networks.

The following examples pertain to further embodiments.

Example 1 may include a device comprising processing circuitry coupled to storage, the processing circuitry configured to: receive a prompt to initiate onboarding of the device to a framework; initiate a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework; receive a list of available identity providers (IdPs) configured for the framework; receive a selection of an IdP of the list of available IdPs; establish a connection

between the IdP and the device; receive an IdP-generated user-specific profile; and connect to the framework using the user-specific profile.

Example 2 may include the device of example 1 and/or some other example herein, wherein the prompt to initiate onboarding of the device to the framework is received in response to receiving an advertisement associated with the framework from the ANP.

Example 3 may include the device of example 1 and/or some other example herein, wherein the ANP is registered with the framework.

Example 4 may include the device of example 3 and/or some other example herein, wherein the ANP is issued a certificate during registration, and wherein the certificate is used by the device to verify that the ANP is registered with the framework.

Example 5 may include the device of example 1 and/or some other example herein, wherein establishing the connection between the IdP and the device further comprises: establishing a tunnel between a federated onboarding service and the device; and transmitting information from the device to the federated onboarding service, wherein the federated onboarding service is configured to route onboarding of the device to the IdP.

Example 6 may include the device of example 5 and/or some other example herein, wherein the information comprises at least one of: IdP-specific credentials, terms and conditions, or user account validation or creation.

Example 7 may include the device of example 1 and/or some other example herein, wherein the user-specific profile is used to authenticate the device prior to the device connecting to the framework.

Example 8 may include the device of example 1 and/or some other example herein, wherein the list of available IdPs configured for the framework comprises at least one profile associated with a user associated with the device.

Example 9 may include the device of example 1 and/or some other example herein, further comprising a transceiver configured to transmit and receive wireless signals.

Example 10 may include the device of example 9 and/or some other example herein, further comprising an antenna coupled to the transceiver to connect to the framework using the user-specific profile.

Example 11 may include a non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors result in performing operations comprising: receiving a prompt to initiate onboarding of the device to a framework; initiating a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework; receiving a list of available identity providers (IdPs) configured for the framework; receiving a selection of an IdP of the list of available IdPs; establishing a connection between the IdP and the device; receiving an IdP-generated user-specific profile; and connecting to the framework using the user-specific profile.

Example 12 may include the non-transitory computer-readable medium of example 11 and/or some other example herein, wherein the prompt to initiate onboarding of the device to the framework is received in response to receiving an advertisement associated with the framework from the ANP.

Example 13 may include the non-transitory computer-readable medium of example 11 and/or some other example herein, wherein the ANP is registered with the framework and is issued a certificate during registration, and wherein

the certificate is used by the device to verify that the ANP is registered with the framework.

Example 14 may include the non-transitory computer-readable medium of example 11 and/or some other example herein, wherein establishing the connection between the IdP and the device further comprises: establishing a tunnel between a federated onboarding service and the device; and transmitting information from the device to the federated onboarding service, wherein the federated onboarding service is configured to route onboarding of the device to the IdP.

Example 15 may include the non-transitory computer-readable medium of example 14 and/or some other example herein, wherein the information comprises at least one of: IdP-specific credentials, terms and conditions, or user account validation or creation.

Example 16 may include the non-transitory computer-readable medium of example 11 and/or some other example herein, wherein the user-specific profile is used to authenticate the device prior to the device connecting to the framework.

Example 17 may include a method comprising: receiving a prompt to initiate onboarding of the device to a framework; initiating a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework; receiving a list of available identity providers (IdPs) configured for the framework; receiving a selection of an IdP of the list of available IdPs; establishing a connection between the IdP and the device; receiving an IdP-generated user-specific profile; and connecting to the framework using the user-specific profile.

Example 18 may include the method of example 17 and/or some other example herein, wherein the prompt to initiate onboarding of the device to the framework is received in response to receiving an advertisement associated with the framework from the ANP.

Example 19 may include the method of example 17 and/or some other example herein, wherein the ANP is registered with the framework and is issued a certificate during registration, and wherein the certificate is used by the device to verify that the ANP is registered with the framework.

Example 20 may include the method of example 17 and/or some other example herein, wherein establishing the connection between the IdP and the device further comprises: establishing a tunnel between a federated onboarding service and the device; and transmitting information from the device to the federated onboarding service, wherein the federated onboarding service is configured to route onboarding of the device to the IdP.

Example 21 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-20, or any other method or process described herein.

Example 22 may include an apparatus comprising logic, modules, and/or circuitry to perform one or more elements of a method described in or related to any of examples 1-20, or any other method or process described herein.

Example 23 may include a method, technique, or process as described in or related to any of examples 1-20, or portions or parts thereof.

Example 24 may include an apparatus comprising: one or more processors and one or more computer readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-20, or portions thereof.

Example 25 may include a method of communicating in a wireless network as shown and described herein.

Example 26 may include a system for providing wireless communication as shown and described herein.

Example 27 may include a device for providing wireless communication as shown and described herein.

Embodiments according to the disclosure are in particular disclosed in the attached claims directed to a method, a storage medium, a device and a computer program product, wherein any feature mentioned in one claim category, e.g., method, can be claimed in another claim category, e.g., system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Certain aspects of the disclosure are described above with reference to block and flow diagrams of systems, methods, apparatuses, and/or computer program products according to various implementations. It will be understood that one or more blocks of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and the flow diagrams, respectively, may be implemented by computer-executable program instructions. Likewise, some blocks of the block diagrams and flow diagrams may not necessarily need to be performed in the order presented, or may not necessarily need to be performed at all, according to some implementations.

These computer-executable program instructions may be loaded onto a special-purpose computer or other particular machine, a processor, or other programmable data processing apparatus to produce a particular machine, such that the instructions that execute on the computer, processor, or other programmable data processing apparatus create means for implementing one or more functions specified in the flow diagram block or blocks. These computer program instructions may also be stored in a computer-readable storage media or memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable storage media produce an article of manufacture including instruction means that implement one or more functions specified in the flow diagram block or

blocks. As an example, certain implementations may provide for a computer program product, comprising a computer-readable storage medium having a computer-readable program code or program instructions implemented therein, said computer-readable program code adapted to be executed to implement one or more functions specified in the flow diagram block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational elements or steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions that execute on the computer or other programmable apparatus provide elements or steps for implementing the functions specified in the flow diagram block or blocks.

Accordingly, blocks of the block diagrams and flow diagrams support combinations of means for performing the specified functions, combinations of elements or steps for performing the specified functions and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and flow diagrams, may be implemented by special-purpose, hardware-based computer systems that perform the specified functions, elements or steps, or combinations of special-purpose hardware and computer instructions.

Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain implementations could include, while other implementations do not include, certain features, elements, and/or operations. Thus, such conditional language is not generally intended to imply that features, elements, and/or operations are in any way required for one or more implementations or that one or more implementations necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or operations are included or are to be performed in any particular implementation.

Many modifications and other implementations of the disclosure set forth herein will be apparent having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific implementations disclosed and that modifications and other implementations are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

What is claimed is:

1. A device, the device comprising processing circuitry coupled to storage, the processing circuitry configured to:
 - receive a prompt to initiate onboarding of the device to a framework;
 - initiate a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework;
 - receive a list of available identity providers (IdPs) configured for the framework;
 - receive a selection of an IdP of the list of available IdPs;
 - establish a connection between the IdP and the device;
 - receive an IdP-generated user-specific profile;
 - validate a federated onboarding service prior to providing user interactions, the user interactions comprising at

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least one of IdP credentials, account validation, account creation, and terms and conditions specific to the selected IdP; and

connect to the framework using the user-specific profile.

2. The device of claim 1, wherein the prompt to initiate onboarding of the device to the framework is received in response to receiving an advertisement associated with the framework from the ANP.

3. The device of claim 1, wherein the ANP is registered with the framework.

4. The device of claim 3, wherein the ANP is issued a certificate during registration, and wherein the certificate is used by the device to verify that the ANP is registered with the framework.

5. The device of claim 1, wherein establishing the connection between the IdP and the device further comprises: establishing a tunnel between a federated onboarding service and the device; and

transmitting information from the device to the federated onboarding service, wherein the federated onboarding service is configured to route onboarding of the device to the IdP.

6. The device of claim 5, wherein the information comprises at least one of: IdP-specific credentials, terms and conditions, or user account validation or creation.

7. The device of claim 1, wherein the user-specific profile is used to authenticate the device prior to the device connecting to the framework.

8. The device of claim 1, wherein the list of available IdPs configured for the framework comprises at least one profile associated with a user associated with the device.

9. The device of claim 1, further comprising a transceiver configured to transmit and receive wireless signals.

10. The device of claim 9, further comprising an antenna coupled to the transceiver to connect to the framework using the user-specific profile.

11. A non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors result in performing operations comprising:

receiving a prompt to initiate onboarding of the device to a framework;

initiating a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework;

receiving a list of available identity providers (IdPs) configured for the framework;

receiving a selection of an IdP of the list of available IdPs; establishing a connection between the IdP and the device; receiving an IdP-generated user-specific profile;

validating a federated onboarding service prior to providing user interactions, the user interactions comprising at least one of IdP credentials, account validation, account creation, and terms and conditions specific to the selected IdP; and

connecting to the framework using the user-specific profile.

12. The non-transitory computer-readable medium of claim 11, wherein the prompt to initiate onboarding of the

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device to the framework is received in response to receiving an advertisement associated with the framework from the ANP.

13. The non-transitory computer-readable medium of claim 11, wherein the ANP is registered with the framework and is issued a certificate during registration, and wherein the certificate is used by the device to verify that the ANP is registered with the framework.

14. The non-transitory computer-readable medium of claim 11, wherein establishing the connection between the IdP and the device further comprises:

establishing a tunnel between a federated onboarding service and the device; and

transmitting information from the device to the federated onboarding service, wherein the federated onboarding service is configured to route onboarding of the device to the IdP.

15. The non-transitory computer-readable medium of claim 14, wherein the information comprises at least one of: IdP-specific credentials, terms and conditions, or user account validation or creation.

16. The non-transitory computer-readable medium of claim 11, wherein the user-specific profile is used to authenticate the device prior to the device connecting to the framework.

17. A method comprising:

receiving a prompt to initiate onboarding of the device to a framework;

initiating a federated onboarding service process at an access network provider (ANP) that is connected to the device, wherein the federated onboarding service process is configured to onboard the device to the framework;

receiving a list of available identity providers (IdPs) configured for the framework;

receiving a selection of an IdP of the list of available IdPs; establishing a connection between the IdP and the device; receiving an IdP-generated user-specific profile;

validating a federated onboarding service prior to providing user interactions, the user interactions comprising at least one of IdP credentials, account validation, account creation, and terms and conditions specific to the selected IdP; and

connecting to the framework using the user-specific profile.

18. The method of claim 17, wherein the prompt to initiate onboarding of the device to the framework is received in response to receiving an advertisement associated with the framework from the ANP.

19. The method of claim 17, wherein the ANP is registered with the framework and is issued a certificate during registration, and wherein the certificate is used by the device to verify that the ANP is registered with the framework.

20. The method of claim 17, wherein establishing the connection between the IdP and the device further comprises:

establishing a tunnel between a federated onboarding service and the device; and

transmitting information from the device to the federated onboarding service, wherein the federated onboarding service is configured to route onboarding of the device to the IdP.

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